
AXIOMS OF PROBABILITY

Contents

- | | | | |
|---|--------------------------|---|--|
| 1 | Introduction | 5 | Sample Spaces Having Equally Likely Outcomes |
| 2 | Sample Space and Events | 6 | Probability as a Continuous Set Function |
| 3 | Axioms of Probability | 7 | Probability as a Measure of Belief |
| 4 | Some Simple Propositions | | |

1 Introduction

In this chapter, we introduce the concept of the probability of an event and then show how probabilities can be computed in certain situations. As a preliminary, however, we need to discuss the concept of the sample space and the events of an experiment.

2 Sample Space and Events

Consider an experiment whose outcome is not predictable with certainty. However, although the outcome of the experiment will not be known in advance, let us suppose that the set of all possible outcomes is known. This set of all possible outcomes of an experiment is known as the *sample space* of the experiment and is denoted by S . Following are some examples:

1. If the outcome of an experiment consists of the determination of the sex of a newborn child, then

$$S = \{g, b\}$$

where the outcome g means that the child is a girl and b that it is a boy.

2. If the outcome of an experiment is the order of finish in a race among the 7 horses having post positions 1, 2, 3, 4, 5, 6, and 7, then

$$S = \{\text{all } 7! \text{ permutations of } (1, 2, 3, 4, 5, 6, 7)\}$$

The outcome $(2, 3, 1, 6, 5, 4, 7)$ means, for instance, that the number 2 horse comes in first, then the number 3 horse, then the number 1 horse, and so on.

3. If the experiment consists of flipping two coins, then the sample space consists of the following four points:

$$S = \{(H, H), (H, T), (T, H), (T, T)\}$$

The outcome will be (H, H) if both coins are heads, (H, T) if the first coin is heads and the second tails, (T, H) if the first is tails and the second heads, and (T, T) if both coins are tails.

4. If the experiment consists of tossing two dice, then the sample space consists of the 36 points

$$S = \{(i, j): i, j = 1, 2, 3, 4, 5, 6\}$$

where the outcome (i, j) is said to occur if i appears on the leftmost die and j on the other die.

5. If the experiment consists of measuring (in hours) the lifetime of a transistor, then the sample space consists of all nonnegative real numbers; that is,

$$S = \{x: 0 \leq x < \infty\}$$

Any subset E of the sample space is known as an *event*. In other words, an event is a set consisting of possible outcomes of the experiment. If the outcome of the experiment is contained in E , then we say that E has occurred. Following are some examples of events.

In the preceding Example 1, if $E = \{g\}$, then E is the event that the child is a girl. Similarly, if $F = \{b\}$, then F is the event that the child is a boy.

In Example 2, if

$$E = \{\text{all outcomes in } S \text{ starting with a 3}\}$$

then E is the event that horse 3 wins the race.

In Example 3, if $E = \{(H, H), (H, T)\}$, then E is the event that a head appears on the first coin.

In Example 4, if $E = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$, then E is the event that the sum of the dice equals 7.

In Example 5, if $E = \{x: 0 \leq x \leq 5\}$, then E is the event that the transistor does not last longer than 5 hours.

For any two events E and F of a sample space S , we define the new event $E \cup F$ to consist of all outcomes that are either in E or in F or in both E and F . That is, the event $E \cup F$ will occur if *either* E or F occurs. For instance, in Example 1, if $E = \{g\}$ is the event that the child is a girl and $F = \{b\}$ is the event that the child is a boy, then

$$E \cup F = \{g, b\}$$

is the whole sample space S . In Example 3, if $E = \{(H, H), (H, T)\}$ is the event that the first coin lands heads, and $F = \{(T, H), (H, H)\}$ is the event that the second coin lands heads, then

$$E \cup F = \{(H, H), (H, T), (T, H)\}$$

is the event that at least one of the coins lands heads and thus will occur provided that both coins do not land tails.

The event $E \cup F$ is called the *union* of the event E and the event F .

Similarly, for any two events E and F , we may also define the new event EF , called the *intersection* of E and F , to consist of all outcomes that are both in E and in F . That is, the event EF (sometimes written $E \cap F$) will occur only if both E and F occur. For instance, in Example 3, if $E = \{(H, H), (H, T), (T, H)\}$ is the event that at least 1 head occurs and $F = \{(H, T), (T, H), (T, T)\}$ is the event that at least 1 tail occurs, then

$$EF = \{(H, T), (T, H)\}$$

is the event that exactly 1 head and 1 tail occur. In Example 4, if $E = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$ is the event that the sum of the dice is 7 and $F = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$ is the event that the sum is 6, then the event EF does not contain any outcomes and hence could not occur. To give such an event a name, we shall refer to it as the null event and denote it by $\emptyset$. (That is, $\emptyset$ refers to the event consisting of no outcomes.) If $EF = \emptyset$, then E and F are said to be *mutually exclusive*.

We define unions and intersections of more than two events in a similar manner. If $E_1, E_2, \dots$ are events, then the union of these events, denoted by $\bigcup_{n=1}^{\infty} E_n$, is defined to be that event that consists of all outcomes that are in E_n for at least one value of $n = 1, 2, \dots$. Similarly, the intersection of the events E_n , denoted by $\bigcap_{n=1}^{\infty} E_n$, is defined to be the event consisting of those outcomes that are in all of the events $E_n, n = 1, 2, \dots$.

Finally, for any event E , we define the new event E^c , referred to as the *complement* of E , to consist of all outcomes in the sample space S that are not in E . That is, E^c will occur if and only if E does not occur. In Example 4, if event $E = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$, then E^c will occur when the sum of the dice does not equal 7. Note that because the experiment must result in some outcome, it follows that $S^c = \emptyset$.

For any two events E and F , if all of the outcomes in E are also in F , then we say that E is *contained* in F , or E is a *subset* of F , and write $E \subset F$ (or equivalently, $F \supset E$, which we sometimes say as F is a *superset* of E). Thus, if $E \subset F$, then the occurrence of E implies the occurrence of F . If $E \subset F$ and $F \subset E$, we say that E and F are equal and write $E = F$.

A graphical representation that is useful for illustrating logical relations among events is the Venn diagram. The sample space S is represented as consisting of all the outcomes in a large rectangle, and the events $E, F, G, \dots$ are represented as consisting of all the outcomes in given circles within the rectangle. Events of interest can then be indicated by shading appropriate regions of the diagram. For instance, in the three Venn diagrams shown in Figure 1, the shaded areas represent,

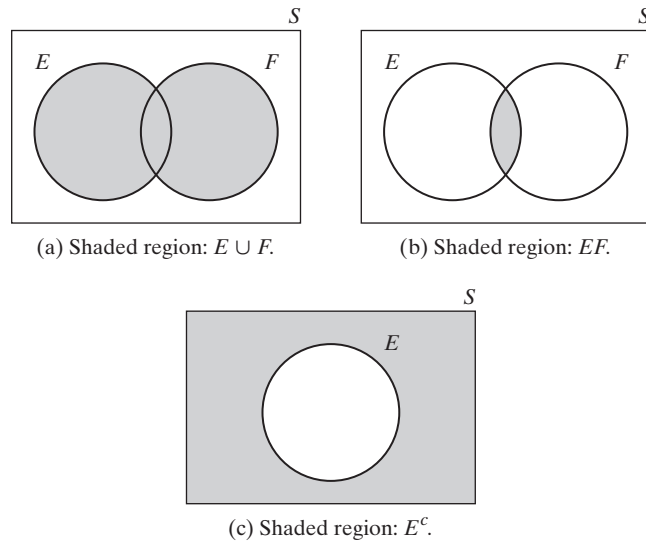


Figure 1 Venn diagrams.

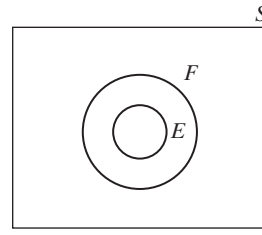


Figure 2 $E \subset F$.

respectively, the events $E \cup F$, EF , and E^c . The Venn diagram in Figure 2 indicates that $E \subset F$.

The operations of forming unions, intersections, and complements of events obey certain rules similar to the rules of algebra. We list a few of these rules:

$$\begin{array}{lll} \text{Commutative laws} & E \cup F = F \cup E & EF = FE \\ \text{Associative laws} & (E \cup F) \cup G = E \cup (F \cup G) & (EF)G = E(FG) \\ \text{Distributive laws} & (E \cup F)G = EG \cup FG & EF \cup G = (E \cup G)(F \cup G) \end{array}$$

These relations are verified by showing that any outcome that is contained in the event on the left side of the equality sign is also contained in the event on the right side, and vice versa. One way of showing this is by means of Venn diagrams. For instance, the distributive law may be verified by the sequence of diagrams in Figure 3.

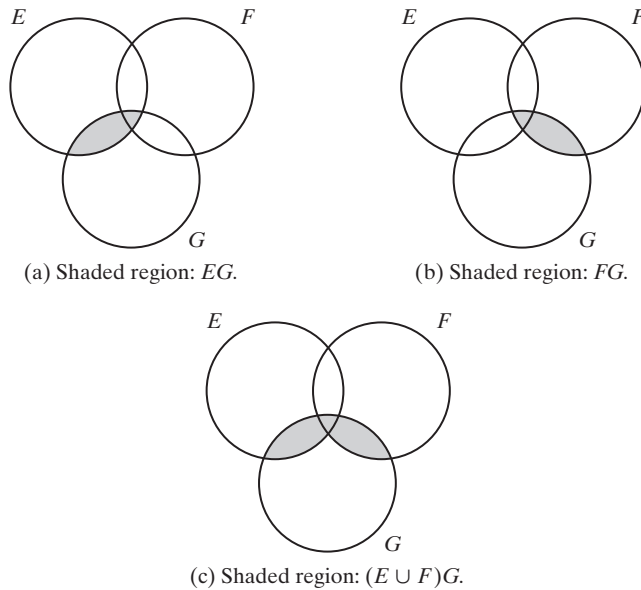


Figure 3 $(E \cup F)G = EG \cup FG$.

The following useful relationships among the three basic operations of forming unions, intersections, and complements are known as *DeMorgan's laws*:

$$\begin{aligned}\left(\bigcup_{i=1}^n E_i\right)^c &= \bigcap_{i=1}^n E_i^c \\ \left(\bigcap_{i=1}^n E_i\right)^c &= \bigcup_{i=1}^n E_i^c\end{aligned}$$

For instance, for two events E and F , DeMorgan's laws state that

$$(E \cup F)^c = E^c F^c \quad \text{and} \quad (EF)^c = E^c \cup F^c$$

which can be easily proven by using Venn diagrams (see Theoretical Exercise 7).

To prove DeMorgan's laws for general n , suppose first that x is an outcome of $\left(\bigcup_{i=1}^n E_i\right)^c$. Then x is not contained in $\bigcup_{i=1}^n E_i$, which means that x is not contained in any of the events $E_i, i = 1, 2, \dots, n$, implying that x is contained in E_i^c for all $i = 1, 2, \dots, n$ and thus is contained in $\bigcap_{i=1}^n E_i^c$. To go the other way, suppose that x is an outcome of $\bigcap_{i=1}^n E_i^c$. Then x is contained in E_i^c for all $i = 1, 2, \dots, n$, which means that x is not contained in E_i for any $i = 1, 2, \dots, n$, implying that x is not contained in $\bigcup_i E_i$, in turn implying that x is contained in $\left(\bigcup_1^n E_i\right)^c$. This proves the first of DeMorgan's laws.

To prove the second of DeMorgan's laws, we use the first law to obtain

$$\left(\bigcup_{i=1}^n E_i^c\right)^c = \bigcap_{i=1}^n (E_i^c)^c$$

which, since $(E^c)^c = E$, is equivalent to

$$\left(\bigcup_1^n E_i^c\right)^c = \bigcap_1^n E_i$$

Taking complements of both sides of the preceding equation yields the result we seek, namely,

$$\bigcup_1^n E_i^c = \left(\bigcap_1^n E_i\right)^c$$

3 Axioms of Probability

One way of defining the probability of an event is in terms of its *relative frequency*. Such a definition usually goes as follows: We suppose that an experiment, whose sample space is S , is repeatedly performed under exactly the same conditions. For each event E of the sample space S , we define $n(E)$ to be the number of times in

the first n repetitions of the experiment that the event E occurs. Then $P(E)$, the probability of the event E , is defined as

$$P(E) = \lim_{n \rightarrow \infty} \frac{n(E)}{n}$$

That is, $P(E)$ is defined as the (limiting) proportion of time that E occurs. It is thus the limiting relative frequency of E .

Although the preceding definition is certainly intuitively pleasing and should always be kept in mind by the reader, it possesses a serious drawback: How do we know that $n(E)/n$ will converge to some constant limiting value that will be the same for each possible sequence of repetitions of the experiment? For example, suppose that the experiment to be repeatedly performed consists of flipping a coin. How do we know that the proportion of heads obtained in the first n flips will converge to some value as n gets large? Also, even if it does converge to some value, how do we know that, if the experiment is repeatedly performed a second time, we shall obtain the same limiting proportion of heads?

Proponents of the relative frequency definition of probability usually answer this objection by stating that the convergence of $n(E)/n$ to a constant limiting value is an assumption, or an *axiom*, of the system. However, to assume that $n(E)/n$ will necessarily converge to some constant value seems to be an extraordinarily complicated assumption. For, although we might indeed hope that such a constant limiting frequency exists, it does not at all seem to be a priori evident that this need be the case. In fact, would it not be more reasonable to assume a set of simpler and more self-evident axioms about probability and then attempt to prove that such a constant limiting frequency does in some sense exist? The latter approach is the modern axiomatic approach to probability theory that we shall adopt in this chapter. In particular, we shall assume that, for each event E in the sample space S , there exists a value $P(E)$, referred to as the probability of E . We shall then assume that all these probabilities satisfy a certain set of axioms, which, we hope the reader will agree, is in accordance with our intuitive notion of probability.

Consider an experiment whose sample space is S . For each event E of the sample space S , we assume that a number $P(E)$ is defined and satisfies the following three axioms:

The three axioms of probability

Axiom 1

$$0 \leq P(E) \leq 1$$

Axiom 2

$$P(S) = 1$$

Axiom 3

For any sequence of mutually exclusive events $E_1, E_2, \dots$ (that is, events for which $E_i E_j = \emptyset$ when $i \neq j$),

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$$

We refer to $P(E)$ as the *probability* of the event E .

Thus, Axiom 1 states that the probability that the outcome of the experiment is an outcome in E is some number between 0 and 1. Axiom 2 states that, with probability 1, the outcome will be a point in the sample space S . Axiom 3 states that, for any sequence of mutually exclusive events, the probability of at least one of these events occurring is just the sum of their respective probabilities.

If we consider a sequence of events $E_1, E_2, \dots$, where $E_1 = S$ and $E_i = \emptyset$ for $i > 1$, then, because the events are mutually exclusive and because $S = \bigcup_{i=1}^{\infty} E_i$, we have, from Axiom 3,

$$P(S) = \sum_{i=1}^{\infty} P(E_i) = P(S) + \sum_{i=2}^{\infty} P(\emptyset)$$

implying that

$$P(\emptyset) = 0$$

That is, the null event has probability 0 of occurring.

Note that it follows that, for any finite sequence of mutually exclusive events $E_1, E_2, \dots, E_n$,

$$P\left(\bigcup_{i=1}^n E_i\right) = \sum_{i=1}^n P(E_i) \quad (3.1)$$

This equation follows from Axiom 3 by defining E_i as the null event for all values of i greater than n . Axiom 3 is equivalent to Equation (3.1) when the sample space is finite. (Why?) However, the added generality of Axiom 3 is necessary when the sample space consists of an infinite number of points.

**Example
3a**

If our experiment consists of tossing a coin and if we assume that a head is as likely to appear as a tail, then we would have

$$P(\{H\}) = P(\{T\}) = \frac{1}{2}$$

On the other hand, if the coin were biased and we believed that a head were twice as likely to appear as a tail, then we would have

$$P(\{H\}) = \frac{2}{3} \quad P(\{T\}) = \frac{1}{3} \quad \blacksquare$$

**Example
3b**

If a die is rolled and we suppose that all six sides are equally likely to appear, then we would have $P(\{1\}) = P(\{2\}) = P(\{3\}) = P(\{4\}) = P(\{5\}) = P(\{6\}) = \frac{1}{6}$. From Axiom 3, it would thus follow that the probability of rolling an even number would equal

$$P(\{2, 4, 6\}) = P(\{2\}) + P(\{4\}) + P(\{6\}) = \frac{1}{2} \quad \blacksquare$$

The assumption of the existence of a set function P , defined on the events of a sample space S and satisfying Axioms 1, 2, and 3, constitutes the modern mathematical approach to probability theory. It is hoped that the reader will agree that the axioms are natural and in accordance with our intuitive concept of probability as related to chance and randomness. Furthermore, using these axioms, we shall be able to prove that if an experiment is repeated over and over again, then, with

probability 1, the proportion of time during which any specific event E occurs will equal $P(E)$. This result is known as the strong law of large numbers. In addition, we present another possible interpretation of probability—as being a measure of belief—in Section 7.

Technical Remark. We have supposed that $P(E)$ is defined for all the events E of the sample space. Actually, when the sample space is an uncountably infinite set, $P(E)$ is defined only for a class of events called *measurable*. However, this restriction need not concern us, as all events of any practical interest are measurable.

4 Some Simple Propositions

In this section, we prove some simple propositions regarding probabilities. We first note that since E and E^c are always mutually exclusive and since $E \cup E^c = S$, we have, by Axioms 2 and 3,

$$1 = P(S) = P(E \cup E^c) = P(E) + P(E^c)$$

Or, equivalently, we have Proposition 4.1.

**Proposition
4.1**

$$P(E^c) = 1 - P(E)$$

In words, Proposition 4.1 states that the probability that an event does not occur is 1 minus the probability that it does occur. For instance, if the probability of obtaining a head on the toss of a coin is $\frac{3}{8}$, then the probability of obtaining a tail must be $\frac{5}{8}$.

Our second proposition states that if the event E is contained in the event F , then the probability of E is no greater than the probability of F .

**Proposition
4.2**

If $E \subset F$, then $P(E) \leq P(F)$.

Proof Since $E \subset F$, it follows that we can express F as

$$F = E \cup E^c F$$

Hence, because E and $E^c F$ are mutually exclusive, we obtain, from Axiom 3,

$$P(F) = P(E) + P(E^c F)$$

which proves the result, since $P(E^c F) \geq 0$. □

Proposition 4.2 tells us, for instance, that the probability of rolling a 1 with a die is less than or equal to the probability of rolling an odd value with the die.

The next proposition gives the relationship between the probability of the union of two events, expressed in terms of the individual probabilities, and the probability of the intersection of the events.

**Proposition
4.3**

$$P(E \cup F) = P(E) + P(F) - P(EF)$$

Proof To derive a formula for $P(E \cup F)$, we first note that $E \cup F$ can be written as the union of the two disjoint events E and $E^c F$. Thus, from Axiom 3, we obtain

$$\begin{aligned} P(E \cup F) &= P(E \cup E^c F) \\ &= P(E) + P(E^c F) \end{aligned}$$

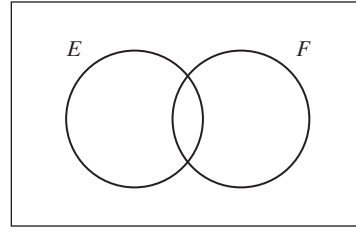


Figure 4 Venn diagram.

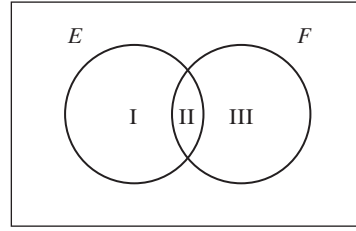


Figure 5 Venn diagram in sections.

Furthermore, since $F = EF \cup E^cF$, we again obtain from Axiom 3

$$P(F) = P(EF) + P(E^cF)$$

or, equivalently,

$$P(E^cF) = P(F) - P(EF)$$

thereby completing the proof. $\square$

Proposition 4.3 could also have been proved by making use of the Venn diagram in Figure 4.

Let us divide $E \cup F$ into three mutually exclusive sections, as shown in Figure 5. In words, section I represents all the points in E that are not in F (that is, EF^c), section II represents all points both in E and in F (that is, EF), and section III represents all points in F that are not in E (that is, E^cF).

From Figure 5, we see that

$$E \cup F = I \cup II \cup III$$

$$E = I \cup II$$

$$F = II \cup III$$

As I, II, and III are mutually exclusive, it follows from Axiom 3 that

$$P(E \cup F) = P(I) + P(II) + P(III)$$

$$P(E) = P(I) + P(II)$$

$$P(F) = P(II) + P(III)$$

which shows that

$$P(E \cup F) = P(E) + P(F) - P(II)$$

and Proposition 4.3 is proved, since $II = EF$.

**Example
4a**

J is taking two books along on her holiday vacation. With probability .5, she will like the first book; with probability .4, she will like the second book; and with probability .3, she will like both books. What is the probability that she likes neither book?

Solution Let B_i denote the event that J likes book $i, i = 1, 2$. Then the probability that she likes at least one of the books is

$$P(B_1 \cup B_2) = P(B_1) + P(B_2) - P(B_1 B_2) = .5 + .4 - .3 = .6$$

Because the event that J likes neither book is the complement of the event that she likes at least one of them, we obtain the result

$$P(B_1^c B_2^c) = P((B_1 \cup B_2)^c) = 1 - P(B_1 \cup B_2) = .4 \quad \blacksquare$$

We may also calculate the probability that any one of the three events E, F , and G occurs, namely,

$$P(E \cup F \cup G) = P[(E \cup F) \cup G]$$

which, by Proposition 4.3, equals

$$P(E \cup F) + P(G) - P[(E \cup F)G]$$

Now, it follows from the distributive law that the events $(E \cup F)G$ and $EG \cup FG$ are equivalent; hence, from the preceding equations, we obtain

$$\begin{aligned} P(E \cup F \cup G) &= P(E) + P(F) - P(EF) + P(G) - P(EG \cup FG) \\ &= P(E) + P(F) - P(EF) + P(G) - P(EG) - P(FG) + P(EGFG) \\ &= P(E) + P(F) + P(G) - P(EF) - P(EG) - P(FG) + P(EFG) \end{aligned}$$

In fact, the following proposition, known as the *inclusion-exclusion identity*, can be proved by mathematical induction:

**Proposition
4.4**

$$\begin{aligned} P(E_1 \cup E_2 \cup \dots \cup E_n) &= \sum_{i=1}^n P(E_i) - \sum_{i_1 < i_2} P(E_{i_1} E_{i_2}) + \dots \\ &\quad + (-1)^{r+1} \sum_{i_1 < i_2 < \dots < i_r} P(E_{i_1} E_{i_2} \dots E_{i_r}) \\ &\quad + \dots + (-1)^{n+1} P(E_1 E_2 \dots E_n) \end{aligned}$$

The summation $\sum_{i_1 < i_2 < \dots < i_r} P(E_{i_1} E_{i_2} \dots E_{i_r})$ is taken over all of the $\binom{n}{r}$ possible subsets of size r of the set $\{1, 2, \dots, n\}$.

In words, Proposition 4.4 states that the probability of the union of n events equals the sum of the probabilities of these events taken one at a time, minus the sum of the probabilities of these events taken two at a time, plus the sum of the probabilities of these events taken three at a time, and so on.

Remarks 1. For a noninductive argument for Proposition 4.4, note first that if an outcome of the sample space is not a member of any of the sets E_i , then its probability does not contribute anything to either side of the equality. Now, suppose that an outcome is in exactly m of the events E_i , where $m > 0$. Then, since it is in $\bigcup_i E_i$, its

probability is counted once in $P\left(\bigcup_i E_i\right)$; also, as this outcome is contained in $\binom{m}{k}$

subsets of the type $E_{i_1} E_{i_2} \cdots E_{i_k}$, its probability is counted

$$\binom{m}{1} - \binom{m}{2} + \binom{m}{3} - \cdots \pm \binom{m}{m}$$

times on the right of the equality sign in Proposition 4.4. Thus, for $m > 0$, we must show that

$$1 = \binom{m}{1} - \binom{m}{2} + \binom{m}{3} - \cdots \pm \binom{m}{m}$$

However, since $1 = \binom{m}{0}$, the preceding equation is equivalent to

$$\sum_{i=0}^m \binom{m}{i} (-1)^i = 0$$

and the latter equation follows from the binomial theorem, since

$$0 = (-1 + 1)^m = \sum_{i=0}^m \binom{m}{i} (-1)^i (1)^{m-i}$$

2. The following is a succinct way of writing the inclusion–exclusion identity:

$$P(\cup_{i=1}^n E_i) = \sum_{r=1}^n (-1)^{r+1} \sum_{i_1 < \cdots < i_r} P(E_{i_1} \cdots E_{i_r})$$

3. In the inclusion–exclusion identity, going out one term results in an upper bound on the probability of the union, going out two terms results in a lower bound on the probability, going out three terms results in an upper bound on the probability, going out four terms results in a lower bound, and so on. That is, for events $E_1, \dots, E_n$, we have

$$P(\cup_{i=1}^n E_i) \leq \sum_{i=1}^n P(E_i) \tag{4.1}$$

$$P(\cup_{i=1}^n E_i) \geq \sum_{i=1}^n P(E_i) - \sum_{j < i} P(E_i E_j) \tag{4.2}$$

$$P(\cup_{i=1}^n E_i) \leq \sum_{i=1}^n P(E_i) - \sum_{j < i} P(E_i E_j) + \sum_{k < j < i} P(E_i E_j E_k) \tag{4.3}$$

and so on. To prove the validity of these bounds, note the identity

$$\cup_{i=1}^n E_i = E_1 \cup E_1^c E_2 \cup E_1^c E_2^c E_3 \cup \cdots \cup E_1^c \cdots E_{n-1}^c E_n$$

That is, at least one of the events E_i occurs if E_1 occurs, or if E_1 does not occur but E_2 does, or if E_1 and E_2 do not occur but E_3 does, and so on. Because the right-hand side is the union of disjoint events, we obtain

$$\begin{aligned} P(\cup_{i=1}^n E_i) &= P(E_1) + P(E_1^c E_2) + P(E_1^c E_2^c E_3) + \cdots + P(E_1^c \cdots E_{n-1}^c E_n) \\ &= P(E_1) + \sum_{i=2}^n P(E_1^c \cdots E_{i-1}^c E_i) \end{aligned} \tag{4.4}$$

Now, let $B_i = E_1^c \cdots E_{i-1}^c = (\cup_{j<i} E_j)^c$ be the event that none of the first $i - 1$ events occurs. Applying the identity

$$P(E_i) = P(B_i E_i) + P(B_i^c E_i)$$

shows that

$$P(E_i) = P(E_1^c \cdots E_{i-1}^c E_i) + P(E_i \cup_{j<i} E_j)$$

or, equivalently,

$$P(E_1^c \cdots E_{i-1}^c E_i) = P(E_i) - P(\cup_{j<i} E_i E_j)$$

Substituting this equation into (4.4) yields

$$P(\cup_{i=1}^n E_i) = \sum_i P(E_i) - \sum_i P(\cup_{j<i} E_i E_j) \quad (4.5)$$

Because probabilities are always nonnegative, Inequality (4.1) follows directly from Equation (4.5). Now, fixing i and applying Inequality (4.1) to $P(\cup_{j<i} E_i E_j)$ yields

$$P(\cup_{j<i} E_i E_j) \leq \sum_{j<i} P(E_i E_j)$$

which, by Equation (4.5), gives Inequality (4.2). Similarly, fixing i and applying Inequality (4.2) to $P(\cup_{j<i} E_i E_j)$ yields

$$\begin{aligned} P(\cup_{j<i} E_i E_j) &\geq \sum_{j<i} P(E_i E_j) - \sum_{k<j<i} P(E_i E_j E_k) \\ &= \sum_{j<i} P(E_i E_j) - \sum_{k<j<i} P(E_i E_j E_k) \end{aligned}$$

which, by Equation (4.5), gives Inequality (4.3). The next inclusion-exclusion inequality is now obtained by fixing i and applying Inequality (4.3) to $P(\cup_{j<i} E_i E_j)$, and so on.

5 Sample Spaces Having Equally Likely Outcomes

In many experiments, it is natural to assume that all outcomes in the sample space are equally likely to occur. That is, consider an experiment whose sample space S is a finite set, say, $S = \{1, 2, \dots, N\}$. Then, it is often natural to assume that

$$P(\{1\}) = P(\{2\}) = \cdots = P(\{N\})$$

which implies, from Axioms 2 and 3 (why?), that

$$P(\{i\}) = \frac{1}{N} \quad i = 1, 2, \dots, N$$

From this equation, it follows from Axiom 3 that, for any event E ,

$$P(E) = \frac{\text{number of outcomes in } E}{\text{number of outcomes in } S}$$

In words, if we assume that all outcomes of an experiment are equally likely to occur, then the probability of any event E equals the proportion of outcomes in the sample space that are contained in E .

**Example
5a**

If two dice are rolled, what is the probability that the sum of the upturned faces will equal 7?

Solution We shall solve this problem under the assumption that all of the 36 possible outcomes are equally likely. Since there are 6 possible outcomes—namely, (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), and (6, 1)—that result in the sum of the dice being equal to 7, the desired probability is $\frac{6}{36} = \frac{1}{6}$. ■

**Example
5b**

If 3 balls are “randomly drawn” from a bowl containing 6 white and 5 black balls, what is the probability that one of the balls is white and the other two black?

Solution If we regard the balls as being distinguishable and the order in which they are selected as being relevant, then the sample space consists of $11 \cdot 10 \cdot 9 = 990$ outcomes. Furthermore, there are $6 \cdot 5 \cdot 4 = 120$ outcomes in which the first ball selected is white and the other two are black; $5 \cdot 6 \cdot 4 = 120$ outcomes in which the first is black, the second is white, and the third is black; and $5 \cdot 4 \cdot 6 = 120$ in which the first two are black and the third is white. Hence, assuming that “randomly drawn” means that each outcome in the sample space is equally likely to occur, we see that the desired probability is

$$\frac{120 + 120 + 120}{990} = \frac{4}{11}$$

This problem could also have been solved by regarding the outcome of the experiment as the unordered set of drawn balls. From this point of view, there are $\binom{11}{3} = 165$ outcomes in the sample space. Now, each set of 3 balls corresponds to $3!$ outcomes when the order of selection is noted. As a result, if all outcomes are assumed equally likely when the order of selection is noted, then it follows that they remain equally likely when the outcome is taken to be the unordered set of selected balls. Hence, using the latter representation of the experiment, we see that the desired probability is

$$\frac{\binom{6}{1}\binom{5}{2}}{\binom{11}{3}} = \frac{4}{11}$$

which, of course, agrees with the answer obtained previously in the chapter. ■

When the experiment consists of a random selection of k items from a set of n items, we have the flexibility of either letting the outcome of the experiment be the ordered selection of the k items or letting it be the unordered set of items selected. In the former case, we would assume that each new selection is equally likely to be any of the so far unselected items of the set, and in the latter case, we would assume that all $\binom{n}{k}$ possible subsets of k items are equally likely to be the set selected. For instance, suppose 5 people are to be randomly selected from a group of 20 individuals consisting of 10 married couples, and we want to determine $P(N)$, the probability that the 5 chosen are all unrelated. (That is, no two are married to each other.) If we regard the sample space as the set of 5 people chosen, then there are $\binom{20}{5}$ equally likely outcomes. An outcome that does not contain a married couple can be thought of as being the result of a six-stage experiment: In the first stage, 5 of the 10 couples to have a member in the group are chosen; in the next 5 stages, 1 of the 2 members of

each of these couples is selected. Thus, there are $\binom{10}{5}2^5$ possible outcomes in which the 5 members selected are unrelated, yielding the desired probability of

$$P(N) = \frac{\binom{10}{5}2^5}{\binom{20}{5}}$$

In contrast, we could let the outcome of the experiment be the *ordered* selection of the 5 individuals. In this setting, there are $20 \cdot 19 \cdot 18 \cdot 17 \cdot 16$ equally likely outcomes, of which $20 \cdot 18 \cdot 16 \cdot 14 \cdot 12$ outcomes result in a group of 5 unrelated individuals, yielding the result

$$P(N) = \frac{20 \cdot 18 \cdot 16 \cdot 14 \cdot 12}{20 \cdot 19 \cdot 18 \cdot 17 \cdot 16}$$

We leave it for the reader to verify that the two answers are identical.

Example 5c

A committee of 5 is to be selected from a group of 6 men and 9 women. If the selection is made randomly, what is the probability that the committee consists of 3 men and 2 women?

Solution Because each of the $\binom{15}{5}$ possible committees is equally likely to be selected, the desired probability is

$$\frac{\binom{6}{3}\binom{9}{2}}{\binom{15}{5}} = \frac{240}{1001} \quad \blacksquare$$

Example 5d

An urn contains n balls, one of which is special. If k of these balls are withdrawn one at a time, with each selection being equally likely to be any of the balls that remain at the time, what is the probability that the special ball is chosen?

Solution Since all of the balls are treated in an identical manner, it follows that the set of k balls selected is equally likely to be any of the $\binom{n}{k}$ sets of k balls. Therefore,

$$P\{\text{special ball is selected}\} = \frac{\binom{1}{1}\binom{n-1}{k-1}}{\binom{n}{k}} = \frac{k}{n}$$

We could also have obtained this result by letting A_i denote the event that the special ball is the i th ball to be chosen, $i = 1, \dots, k$. Then, since each one of the n balls is equally likely to be the i th ball chosen, it follows that $P(A_i) = 1/n$. Hence, because these events are clearly mutually exclusive, we have

$$P\{\text{special ball is selected}\} = P\left(\bigcup_{i=1}^k A_i\right) = \sum_{i=1}^k P(A_i) = \frac{k}{n}$$

We could also have argued that $P(A_i) = 1/n$, by noting that there are $n(n-1)\cdots(n-k+1) = n!/(n-k)!$ equally likely outcomes of the experiment, of which

$(n - 1)(n - 2) \cdots (n - i + 1)(1)(n - i) \cdots (n - k + 1) = (n - 1)!/(n - k)!$ result in the special ball being the i th one chosen. From this reasoning, it follows that

$$P(A_i) = \frac{(n - 1)!}{n!} = \frac{1}{n} \quad \blacksquare$$

**Example
5e**

Suppose that $n + m$ balls, of which n are red and m are blue, are arranged in a linear order in such a way that all $(n + m)!$ possible orderings are equally likely. If we record the result of this experiment by listing only the colors of the successive balls, show that all the possible results remain equally likely.

Solution Consider any one of the $(n + m)!$ possible orderings, and note that any permutation of the red balls among themselves and of the blue balls among themselves does not change the sequence of colors. As a result, every ordering of colorings corresponds to $n! m!$ different orderings of the $n + m$ balls, so every ordering of the colors has probability $\frac{n!m!}{(n+m)!}$ of occurring.

For example, suppose that there are 2 red balls, numbered r_1, r_2 , and 2 blue balls, numbered b_1, b_2 . Then, of the $4!$ possible orderings, there will be $2! 2!$ orderings that result in any specified color combination. For instance, the following orderings result in the successive balls alternating in color, with a red ball first:

$$r_1, b_1, r_2, b_2 \quad r_1, b_2, r_2, b_1 \quad r_2, b_1, r_1, b_2 \quad r_2, b_2, r_1, b_1$$

Therefore, each of the possible orderings of the colors has probability $\frac{4}{24} = \frac{1}{6}$ of occurring. $\blacksquare$

**Example
5f**

A poker hand consists of 5 cards. If the cards have distinct consecutive values and are not all of the same suit, we say that the hand is a straight. For instance, a hand consisting of the five of spades, six of spades, seven of spades, eight of spades, and nine of hearts is a straight. What is the probability that one is dealt a straight?

Solution We start by assuming that all $\binom{52}{5}$ possible poker hands are equally likely. To determine the number of outcomes that are straights, let us first determine the number of possible outcomes for which the poker hand consists of an ace, two, three, four, and five (the suits being irrelevant). Since the ace can be any 1 of the 4 possible aces, and similarly for the two, three, four, and five, it follows that there are 4^5 outcomes leading to exactly one ace, two, three, four, and five. Hence, since in 4 of these outcomes all the cards will be of the same suit (such a hand is called a straight flush), it follows that there are $4^5 - 4$ hands that make up a straight of the form ace, two, three, four, and five. Similarly, there are $4^5 - 4$ hands that make up a straight of the form ten, jack, queen, king, and ace. Thus, there are $10(4^5 - 4)$ hands that are straights, and it follows that the desired probability is

$$\frac{10(4^5 - 4)}{\binom{52}{5}} \approx .0039 \quad \blacksquare$$

**Example
5g**

A 5-card poker hand is said to be a full house if it consists of 3 cards of the same denomination and 2 other cards of the same denomination (of course, different from the first denomination). Thus, a full house is three of a kind plus a pair. What is the probability that one is dealt a full house?

Solution Again, we assume that all $\binom{52}{5}$ possible hands are equally likely. To determine the number of possible full houses, we first note that there are $\binom{4}{2}\binom{4}{3}$ different combinations of, say, 2 tens and 3 jacks. Because there are 13 different choices for the kind of pair and, after a pair has been chosen, there are 12 other choices for the denomination of the remaining 3 cards, it follows that the probability of a full house is

$$\frac{13 \cdot 12 \cdot \binom{4}{2} \binom{4}{3}}{\binom{52}{5}} \approx .0014 \quad \blacksquare$$

Example 5h

In the game of bridge, the entire deck of 52 cards is dealt out to 4 players. What is the probability that

- (a) one of the players receives all 13 spades;
- (b) each player receives 1 ace?

Solution (a) Letting E_i be the event that hand i has all 13 spades, then

$$P(E_i) = \frac{1}{\binom{52}{13}}, \quad i = 1, 2, 3, 4$$

Because the events E_i , $i = 1, 2, 3, 4$, are mutually exclusive, the probability that one of the hands is dealt all 13 spades is

$$P(\cup_{i=1}^4 E_i) = \sum_{i=1}^4 P(E_i) = 4/\binom{52}{13} \approx 6.3 \times 10^{-12}$$

(b) To determine the number of outcomes in which each of the distinct players receives exactly 1 ace, put aside the aces and note that there are $\binom{48}{12, 12, 12, 12}$ possible divisions of the other 48 cards when each player is to receive 12. Because there are $4!$ ways of dividing the 4 aces so that each player receives 1, we see that the number of possible outcomes in which each player receives exactly 1 ace is $4! \binom{48}{12, 12, 12, 12}$.

As there are $\binom{52}{13, 13, 13, 13}$ possible hands, the desired probability is thus

$$\frac{4! \binom{48}{12, 12, 12, 12}}{\binom{52}{13, 13, 13, 13}} \approx .1055 \quad \blacksquare$$

Some results in probability are quite surprising when initially encountered. Our next two examples illustrate this phenomenon.

Example
5i

If n people are present in a room, what is the probability that no two of them celebrate their birthday on the same day of the year? How large need n be so that this probability is less than $\frac{1}{2}$?

Solution As each person can celebrate his or her birthday on any one of 365 days, there are a total of $(365)^n$ possible outcomes. (We are ignoring the possibility of someone having been born on February 29.) Assuming that each outcome is equally likely, we see that the desired probability is $(365)(364)(363) \dots (365 - n + 1)/(365)^n$. It is a rather surprising fact that when $n \geq 23$, this probability is less than $\frac{1}{2}$. That is, if there are 23 or more people in a room, then the probability that at least two of them have the same birthday exceeds $\frac{1}{2}$. Many people are initially surprised by this result, since 23 seems so small in relation to 365, the number of days of the year. However, every pair of individuals has probability $\frac{365}{(365)^2} = \frac{1}{365}$ of having the same birthday, and in a group of 23 people, there are $\binom{23}{2} = 253$ different pairs of individuals. Looked at this way, the result no longer seems so surprising.

When there are 50 people in the room, the probability that at least two share the same birthday is approximately .970, and with 100 persons in the room, the odds are better than 3,000,000:1. (That is, the probability is greater than $\frac{3 \times 10^6}{3 \times 10^6 + 1}$ that at least two people have the same birthday.) ■

Example
5j

A deck of 52 playing cards is shuffled, and the cards are turned up one at a time until the first ace appears. Is the next card—that is, the card following the first ace—more likely to be the ace of spades or the two of clubs?

Solution To determine the probability that the card following the first ace is the ace of spades, we need to calculate how many of the $(52)!$ possible orderings of the cards have the ace of spades immediately following the first ace. To begin, note that each ordering of the 52 cards can be obtained by first ordering the 51 cards different from the ace of spades and then inserting the ace of spades into that ordering. Furthermore, for each of the $(51)!$ orderings of the other cards, there is only one place where the ace of spades can be placed so that it follows the first ace. For instance, if the ordering of the other 51 cards is

$$4c, 6h, Jd, 5s, Ac, 7d, \dots, Kh$$

then the only insertion of the ace of spades into this ordering that results in its following the first ace is

$$4c, 6h, Jd, 5s, Ac, As, 7d, \dots, Kh$$

Therefore, there are $(51)!$ orderings that result in the ace of spades following the first ace, so

$$P\{\text{the ace of spades follows the first ace}\} = \frac{(51)!}{(52)!} = \frac{1}{52}$$

In fact, by exactly the same argument, it follows that the probability that the two of clubs (or any other specified card) follows the first ace is also $\frac{1}{52}$. In other words, each of the 52 cards of the deck is equally likely to be the one that follows the first ace!

Many people find this result rather surprising. Indeed, a common reaction is to suppose initially that it is more likely that the two of clubs (rather than the ace of

spades) follows the first ace, since that first ace might itself be the ace of spades. This reaction is often followed by the realization that the two of clubs might itself appear before the first ace, thus negating its chance of immediately following the first ace. However, as there is one chance in four that the ace of spades will be the first ace (because all 4 aces are equally likely to be first) and only one chance in five that the two of clubs will appear before the first ace (because each of the set of 5 cards consisting of the two of clubs and the 4 aces is equally likely to be the first of this set to appear), it again appears that the two of clubs is more likely. However, this is not the case, and our more complete analysis shows that they are equally likely. ■

Example
5k

A football team consists of 20 offensive and 20 defensive players. The players are to be paired in groups of 2 for the purpose of determining roommates. If the pairing is done at random, what is the probability that there are no offensive–defensive roommate pairs? What is the probability that there are $2i$ offensive–defensive roommate pairs, $i = 1, 2, \dots, 10$?

Solution There are

$$\binom{40}{2, 2, \dots, 2} = \frac{(40)!}{(2!)^{20}}$$

ways of dividing the 40 players into 20 *ordered* pairs of two each. (That is, there are $(40)!/2^{20}$ ways of dividing the players into a *first* pair, a *second* pair, and so on.) Hence, there are $(40)!/2^{20}(20)!$ ways of dividing the players into (unordered) pairs of 2 each. Furthermore, since a division will result in no offensive–defensive pairs if the offensive (and defensive) players are paired among themselves, it follows that there are $[(20)!/2^{10}(10)!]^2$ such divisions. Hence, the probability of no offensive–defensive roommate pairs, call it P_0 , is given by

$$P_0 = \frac{\left(\frac{(20)!}{2^{10}(10)!} \right)^2}{\frac{(40)!}{2^{20}(20)!}} = \frac{[(20)!]^3}{[(10)!]^2(40)!}$$

To determine P_{2i} , the probability that there are $2i$ offensive–defensive pairs, we first note that there are $\binom{20}{2i}^2$ ways of selecting the $2i$ offensive players and the $2i$ defensive players who are to be in the offensive–defensive pairs. These $4i$ players can then be paired up into $(2i)!$ possible offensive–defensive pairs. (This is so because the first offensive player can be paired with any of the $2i$ defensive players, the second offensive player with any of the remaining $2i - 1$ defensive players, and so on.) As the remaining $20 - 2i$ offensive (and defensive) players must be paired among themselves, it follows that there are

$$\binom{20}{2i}^2 (2i)! \left[\frac{(20 - 2i)!}{2^{10-i}(10 - i)!} \right]^2$$

divisions that lead to $2i$ offensive–defensive pairs. Hence,

$$P_{2i} = \frac{\binom{20}{2i}^2 (2i)! \left[\frac{(20 - 2i)!}{2^{10-i}(10 - i)!} \right]^2}{\frac{(40)!}{2^{20}(20)!}} \quad i = 0, 1, \dots, 10$$

The $P_{2i}, i = 0, 1, \dots, 10$, can now be computed, or they can be approximated by making use of a result of Stirling, which shows that $n!$ can be approximated by $n^{n+1/2}e^{-n}\sqrt{2\pi}$. For instance, we obtain

$$P_0 \approx 1.3403 \times 10^{-6}$$

$$P_{10} \approx .345861$$

$$P_{20} \approx 7.6068 \times 10^{-6}$$

■

Our next three examples illustrate the usefulness of the inclusion–exclusion identity (Proposition 4.4). In Example 5l, the introduction of probability enables us to obtain a quick solution to a counting problem.

**Example
5l**

A total of 36 members of a club play tennis, 28 play squash, and 18 play badminton. Furthermore, 22 of the members play both tennis and squash, 12 play both tennis and badminton, 9 play both squash and badminton, and 4 play all three sports. How many members of this club play at least one of three sports?

Solution Let N denote the number of members of the club, and introduce probability by assuming that a member of the club is randomly selected. If, for any subset C of members of the club, we let $P(C)$ denote the probability that the selected member is contained in C , then

$$P(C) = \frac{\text{number of members in } C}{N}$$

Now, with T being the set of members that plays tennis, S being the set that plays squash, and B being the set that plays badminton, we have, from Proposition 4.4,

$$\begin{aligned} P(T \cup S \cup B) &= P(T) + P(S) + P(B) - P(TS) - P(TB) - P(SB) + P(TSB) \\ &= \frac{36 + 28 + 18 - 22 - 12 - 9 + 4}{N} \\ &= \frac{43}{N} \end{aligned}$$

Hence, we can conclude that 43 members play at least one of the sports. ■

The next example in this section not only possesses the virtue of giving rise to a somewhat surprising answer, but is also of theoretical interest.

**Example
5m**

The matching problem

Suppose that each of N men at a party throws his hat into the center of the room. The hats are first mixed up, and then each man randomly selects a hat. What is the probability that none of the men selects his own hat?

Solution We first calculate the complementary probability of at least one man selecting his own hat. Let us denote by $E_i, i = 1, 2, \dots, N$ the event that the i th man selects his own hat. Now, by Proposition 4.4, $P\left(\bigcup_{i=1}^N E_i\right)$, the probability that at least one of the men selects his own hat is given by

$$\begin{aligned} P\left(\bigcup_{i=1}^N E_i\right) &= \sum_{i=1}^N P(E_i) - \sum_{i_1 < i_2} P(E_{i_1}E_{i_2}) + \dots \\ &\quad + (-1)^{n+1} \sum_{i_1 < i_2 < \dots < i_n} P(E_{i_1}E_{i_2} \dots E_{i_n}) \\ &\quad + \dots + (-1)^{N+1} P(E_1E_2 \dots E_N) \end{aligned}$$

If we regard the outcome of this experiment as a vector of N numbers, where the i th element is the number of the hat drawn by the i th man, then there are $N!$ possible outcomes. [The outcome $(1, 2, 3, \dots, N)$ means, for example, that each man selects his own hat.] Furthermore, $E_{i_1} E_{i_2} \dots E_{i_n}$, the event that each of the n men $i_1, i_2, \dots, i_n$ selects his own hat, can occur in any of $(N - n)(N - n - 1) \dots 3 \cdot 2 \cdot 1 = (N - n)!$ possible ways; for, of the remaining $N - n$ men, the first can select any of $N - n$ hats, the second can then select any of $N - n - 1$ hats, and so on. Hence, assuming that all $N!$ possible outcomes are equally likely, we see that

$$P(E_{i_1} E_{i_2} \dots E_{i_n}) = \frac{(N - n)!}{N!}$$

Also, as there are $\binom{N}{n}$ terms in $\sum_{i_1 < i_2 < \dots < i_n} P(E_{i_1} E_{i_2} \dots E_{i_n})$, it follows that

$$\sum_{i_1 < i_2 < \dots < i_n} P(E_{i_1} E_{i_2} \dots E_{i_n}) = \frac{N!(N - n)!}{(N - n)!n!N!} = \frac{1}{n!}$$

Thus,

$$P\left(\bigcup_{i=1}^N E_i\right) = 1 - \frac{1}{2!} + \frac{1}{3!} - \dots + (-1)^{N+1} \frac{1}{N!}$$

Hence, the probability that none of the men selects his own hat is

$$1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \dots + \frac{(-1)^N}{N!} = \sum_{i=0}^N (-1)^i / i!$$

Upon letting $x = -1$ in the identity $e^x = \sum_{i=0}^{\infty} x^i / i!$ the preceding probability when N large is seen to be approximately equal to $e^{-1} \approx .3679$. In other words, for N large, the probability that none of the men selects his own hat is approximately .37. (How many readers would have incorrectly thought that this probability would go to 1 as $N \rightarrow \infty$?) ■

For another illustration of the usefulness of Proposition 4.4, consider the following example.

Example 5n

Compute the probability that if 10 married couples are seated at random at a round table, then no wife sits next to her husband.

Solution If we let $E_i, i = 1, 2, \dots, 10$ denote the event that the i th couple sit next to each other, it follows that the desired probability is $1 - P\left(\bigcup_{i=1}^{10} E_i\right)$. Now, from Proposition 4.4,

$$\begin{aligned} P\left(\bigcup_{i=1}^{10} E_i\right) &= \sum_{i=1}^{10} P(E_i) - \dots + (-1)^{n+1} \sum_{i_1 < i_2 < \dots < i_n} P(E_{i_1} E_{i_2} \dots E_{i_n}) \\ &\quad + \dots - P(E_1 E_2 \dots E_{10}) \end{aligned}$$

To compute $P(E_{i_1} E_{i_2} \dots E_{i_n})$, we first note that there are $19!$ ways of arranging 20 people around a round table. (Why?) The number of arrangements that result in a specified set of n men sitting next to their wives can most easily be obtained by first

thinking of each of the n married couples as being single entities. If this were the case, then we would need to arrange $20 - n$ entities around a round table, and there are clearly $(20 - n - 1)!$ such arrangements. Finally, since each of the n married couples can be arranged next to each other in one of two possible ways, it follows that there are $2^n(20 - n - 1)!$ arrangements that result in a specified set of n men each sitting next to their wives. Therefore,

$$P(E_{i_1} E_{i_2} \cdots E_{i_n}) = \frac{2^n(19 - n)!}{(19)!}$$

Thus, from Proposition 4.4, we obtain that the probability that at least one married couple sits together, namely,

$$\binom{10}{1} 2^1 \frac{(18)!}{(19)!} - \binom{10}{2} 2^2 \frac{(17)!}{(19)!} + \binom{10}{3} 2^3 \frac{(16)!}{(19)!} - \cdots - \binom{10}{10} 2^{10} \frac{9!}{(19)!} \approx .6605$$

and the desired probability is approximately .3395. ■

*Example 50 Runs

Consider an athletic team that had just finished its season with a final record of n wins and m losses. By examining the sequence of wins and losses, we are hoping to determine whether the team had stretches of games in which it was more likely to win than at other times. One way to gain some insight into this question is to count the number of runs of wins and then see how likely that result would be when all $(n + m)!/(n! m!)$ orderings of the n wins and m losses are assumed equally likely. By a run of wins, we mean a consecutive sequence of wins. For instance, if $n = 10, m = 6$, and the sequence of outcomes was $WWLLWWLWLLLWWWW$, then there would be 4 runs of wins—the first run being of size 2, the second of size 3, the third of size 1, and the fourth of size 4.

Suppose now that a team has n wins and m losses. Assuming that all $(n + m)!/(n! m!) = \binom{n + m}{n}$ orderings are equally likely, let us determine the probability that there will be exactly r runs of wins. To do so, consider first any vector of positive integers $x_1, x_2, \dots, x_r$ with $x_1 + \cdots + x_r = n$, and let us see how many outcomes result in r runs of wins in which the i th run is of size $x_i, i = 1, \dots, r$. For any such outcome, if we let y_1 denote the number of losses before the first run of wins, y_2 the number of losses between the first 2 runs of wins, $\dots, y_{r+1}$ the number of losses after the last run of wins, then the y_i satisfy

$$y_1 + y_2 + \cdots + y_{r+1} = m \quad y_1 \geq 0, y_{r+1} \geq 0, y_i > 0, i = 2, \dots, r$$

and the outcome can be represented schematically as

$$\underbrace{LL \dots L}_{y_1} \underbrace{WW \dots W}_{x_1} \underbrace{L \dots L}_{y_2} \underbrace{WW \dots W}_{x_2} \cdots \underbrace{WW}_{x_r} \underbrace{L \dots L}_{y_{r+1}}$$

Hence, the number of outcomes that result in r runs of wins—the i th of size $x_i, i = 1, \dots, r$ —is equal to the number of integers $y_1, \dots, y_{r+1}$ that satisfy the foregoing, or, equivalently, to the number of positive integers

$$\bar{y}_1 = y_1 + 1 \quad \bar{y}_i = y_i, i = 2, \dots, r, \quad \bar{y}_{r+1} = y_{r+1} + 1$$

that satisfy

$$\bar{y}_1 + \bar{y}_2 + \cdots + \bar{y}_{r+1} = m + 2$$

Generally, there are $\binom{m+1}{r}$ such outcomes. Hence, the total number of outcomes that result in r runs of wins is $\binom{m+1}{r}$, multiplied by the number of positive integral solutions of $x_1 + \dots + x_r = n$. Thus, there are $\binom{m+1}{r} \binom{n-1}{r-1}$ outcomes resulting in r runs of wins. As there are $\binom{n+m}{n}$ equally likely outcomes, it follows that

$$P(\{r \text{ runs of wins}\}) = \frac{\binom{m+1}{r} \binom{n-1}{r-1}}{\binom{m+n}{n}} \quad r \geq 1$$

For example, if $n = 8$ and $m = 6$, then the probability of 7 runs is $\binom{7}{7} \binom{7}{6} / \binom{14}{8} = 1/429$ if all $\binom{14}{8}$ outcomes are equally likely. Hence, if the outcome was *WLWLWLWLWWLWLW*, then we might suspect that the team's probability of winning was changing over time. (In particular, the probability that the team wins seems to be quite high when it lost its last game and quite low when it won its last game.) On the other extreme, if the outcome were *WWWWWWWWL L L L L L*, then there would have been only 1 run, and as $P(\{1 \text{ run}\}) = \binom{7}{1} \binom{7}{0} / \binom{14}{8} = 1/429$, it would thus again seem unlikely that the team's probability of winning remained unchanged over its 14 games. ■

*6 Probability as a Continuous Set Function

A sequence of events $\{E_n, n \geq 1\}$ is said to be an increasing sequence if

$$E_1 \subset E_2 \subset \dots \subset E_n \subset E_{n+1} \subset \dots$$

whereas it is said to be a decreasing sequence if

$$E_1 \supset E_2 \supset \dots \supset E_n \supset E_{n+1} \supset \dots$$

If $\{E_n, n \geq 1\}$ is an increasing sequence of events, then we define a new event, denoted by $\lim_{n \rightarrow \infty} E_n$, by

$$\lim_{n \rightarrow \infty} E_n = \bigcup_{i=1}^{\infty} E_i$$

Similarly, if $\{E_n, n \geq 1\}$ is a decreasing sequence of events, we define $\lim_{n \rightarrow \infty} E_n$, by

$$\lim_{n \rightarrow \infty} E_n = \bigcap_{i=1}^{\infty} E_i$$

We now prove the following Proposition 6.1:

Proposition 6.1 If $\{E_n, n \geq 1\}$ is either an increasing or a decreasing sequence of events, then

$$\lim_{n \rightarrow \infty} P(E_n) = P(\lim_{n \rightarrow \infty} E_n)$$

Proof Suppose, first, that $\{E_n, n \geq 1\}$ is an increasing sequence, and define the events $F_n, n \geq 1$, by

$$F_1 = E_1$$

$$F_n = E_n \left(\bigcup_{i=1}^{n-1} E_i \right)^c = E_n E_{n-1}^c \quad n > 1$$

where we have used the fact that $\bigcup_{i=1}^{n-1} E_i = E_{n-1}$, since the events are increasing. In words, F_n consists of those outcomes in E_n that are not in any of the earlier $E_i, i < n$. It is easy to verify that the F_n are mutually exclusive events such that

$$\bigcup_{i=1}^{\infty} F_i = \bigcup_{i=1}^{\infty} E_i \quad \text{and} \quad \bigcup_{i=1}^n F_i = \bigcup_{i=1}^n E_i \quad \text{for all } n \geq 1$$

Thus,

$$\begin{aligned} P\left(\bigcup_{i=1}^{\infty} E_i\right) &= P\left(\bigcup_{i=1}^{\infty} F_i\right) \\ &= \sum_{i=1}^{\infty} P(F_i) \quad (\text{by Axiom 3}) \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n P(F_i) \\ &= \lim_{n \rightarrow \infty} P\left(\bigcup_{i=1}^n F_i\right) \\ &= \lim_{n \rightarrow \infty} P\left(\bigcup_{i=1}^n E_i\right) \\ &= \lim_{n \rightarrow \infty} P(E_n) \end{aligned}$$

which proves the result when $\{E_n, n \geq 1\}$ is increasing.

If $\{E_n, n \geq 1\}$ is a decreasing sequence, then $\{E_n^c, n \geq 1\}$ is an increasing sequence; hence, from the preceding equations,

$$P\left(\bigcup_{i=1}^{\infty} E_i^c\right) = \lim_{n \rightarrow \infty} P(E_n^c)$$

However, because $\bigcup_{i=1}^{\infty} E_i^c = \left(\bigcap_{i=1}^{\infty} E_i\right)^c$, it follows that

$$P\left(\left(\bigcap_{i=1}^{\infty} E_i\right)^c\right) = \lim_{n \rightarrow \infty} P(E_n^c)$$

or, equivalently,

$$1 - P\left(\bigcap_{i=1}^{\infty} E_i\right) = \lim_{n \rightarrow \infty} [1 - P(E_n)] = 1 - \lim_{n \rightarrow \infty} P(E_n)$$

or

$$P\left(\bigcap_{i=1}^{\infty} E_i\right) = \lim_{n \rightarrow \infty} P(E_n)$$

which proves the result. $\square$

Example 6a

Probability and a “paradox”

Suppose that we possess an infinitely large urn and an infinite collection of balls labeled ball number 1, number 2, number 3, and so on. Consider an experiment performed as follows: At 1 minute to 12 P.M., balls numbered 1 through 10 are placed in the urn and ball number 10 is withdrawn. (Assume that the withdrawal takes no time.) At $\frac{1}{2}$ minute to 12 P.M., balls numbered 11 through 20 are placed in the urn and ball number 20 is withdrawn. At $\frac{1}{4}$ minute to 12 P.M., balls numbered 21 through 30 are placed in the urn and ball number 30 is withdrawn. At $\frac{1}{8}$ minute to 12 P.M., and so on. The question of interest is, How many balls are in the urn at 12 P.M.?

The answer to this question is clearly that there is an infinite number of balls in the urn at 12 P.M., since any ball whose number is not of the form $10n$, $n \geq 1$, will have been placed in the urn and will not have been withdrawn before 12 P.M. Hence, the problem is solved when the experiment is performed as described.

However, let us now change the experiment and suppose that at 1 minute to 12 P.M., balls numbered 1 through 10 are placed in the urn and ball number 1 is withdrawn; at $\frac{1}{2}$ minute to 12 P.M., balls numbered 11 through 20 are placed in the urn and ball number 2 is withdrawn; at $\frac{1}{4}$ minute to 12 P.M., balls numbered 21 through 30 are placed in the urn and ball number 3 is withdrawn; at $\frac{1}{8}$ minute to 12 P.M., balls numbered 31 through 40 are placed in the urn and ball number 4 is withdrawn, and so on. For this new experiment, how many balls are in the urn at 12 P.M.?

Surprisingly enough, the answer now is that the urn is *empty* at 12 P.M. For, consider any ball—say, ball number n . At some time prior to 12 P.M. [in particular, at $\left(\frac{1}{2}\right)^{n-1}$ minutes to 12 P.M.], this ball would have been withdrawn from the urn. Hence, for each n , ball number n is not in the urn at 12 P.M.; therefore, the urn must be empty at that time.

Because for all n , the number of balls in the urn after the n th interchange is the same in both variations of the experiment, most people are surprised that the two scenarios produce such different results in the limit. It is important to recognize that the reason the results are different is not because there is an actual *paradox*, or mathematical contradiction, but rather because of the logic of the situation, and also that the surprise results because one’s initial intuition when dealing with infinity is not always correct. (This latter statement is not surprising, for when the theory of the infinite was first developed by the mathematician Georg Cantor in the second half of the nineteenth century, many of the other leading mathematicians of the day called it nonsensical and ridiculed Cantor for making such claims as that the set of all integers and the set of all even integers have the same number of elements.)

We see from the preceding discussion that the manner in which the balls are withdrawn makes a difference. For, in the first case, only balls numbered $10n$, $n \geq 1$,

are ever withdrawn, whereas in the second case all of the balls are eventually withdrawn. Let us now suppose that whenever a ball is to be withdrawn, that ball is randomly selected from among those present. That is, suppose that at 1 minute to 12 P.M. balls numbered 1 through 10 are placed in the urn and a ball is randomly selected and withdrawn, and so on. In this case, how many balls are in the urn at 12 P.M.?

Solution We shall show that, with probability 1, the urn is empty at 12 P.M. Let us first consider ball number 1. Define E_n to be the event that ball number 1 is still in the urn after the first n withdrawals have been made. Clearly,

$$P(E_n) = \frac{9 \cdot 18 \cdot 27 \cdots (9n)}{10 \cdot 19 \cdot 28 \cdots (9n + 1)}$$

[To understand this equation, just note that if ball number 1 is still to be in the urn after the first n withdrawals, the first ball withdrawn can be any one of 9, the second any one of 18 (there are 19 balls in the urn at the time of the second withdrawal, one of which must be ball number 1), and so on. The denominator is similarly obtained.]

Now, the event that ball number 1 is in the urn at 12 P.M. is just the event $\bigcap_{n=1}^{\infty} E_n$. Because the events $E_n, n \geq 1$, are decreasing events, it follows from Proposition 6.1 that

$$\begin{aligned} P\{\text{ball number 1 is in the urn at 12 P.M.}\} \\ &= P\left(\bigcap_{n=1}^{\infty} E_n\right) \\ &= \lim_{n \rightarrow \infty} P(E_n) \\ &= \prod_{n=1}^{\infty} \left(\frac{9n}{9n + 1}\right) \end{aligned}$$

We now show that

$$\prod_{n=1}^{\infty} \frac{9n}{9n + 1} = 0$$

Since

$$\prod_{n=1}^{\infty} \left(\frac{9n}{9n + 1}\right) = \left[\prod_{n=1}^{\infty} \left(\frac{9n + 1}{9n}\right)\right]^{-1}$$

this is equivalent to showing that

$$\prod_{n=1}^{\infty} \left(1 + \frac{1}{9n}\right) = \infty$$

Now, for all $m \geq 1$,

$$\begin{aligned} \prod_{n=1}^{\infty} \left(1 + \frac{1}{9n}\right) &\geq \prod_{n=1}^m \left(1 + \frac{1}{9n}\right) \\ &= \left(1 + \frac{1}{9}\right) \left(1 + \frac{1}{18}\right) \left(1 + \frac{1}{27}\right) \cdots \left(1 + \frac{1}{9m}\right) \\ &> \frac{1}{9} + \frac{1}{18} + \frac{1}{27} + \cdots + \frac{1}{9m} \\ &= \frac{1}{9} \sum_{i=1}^m \frac{1}{i} \end{aligned}$$

Hence, letting $m \rightarrow \infty$ and using the fact that $\sum_{i=1}^{\infty} 1/i = \infty$ yields

$$\prod_{n=1}^{\infty} \left(1 + \frac{1}{9n}\right) = \infty$$

Thus, letting F_i denote the event that ball number i is in the urn at 12 P.M., we have shown that $P(F_1) = 0$. Similarly, we can show that $P(F_i) = 0$ for all i .

(For instance, the same reasoning shows that $P(F_i) = \prod_{n=2}^{\infty} [9n/(9n + 1)]$ for $i = 11, 12, \dots, 20$.) Therefore, the probability that the urn is not empty at 12 P.M., $P\left(\bigcup_{i=1}^{\infty} F_i\right)$, satisfies

$$P\left(\bigcup_{i=1}^{\infty} F_i\right) \leq \sum_{i=1}^{\infty} P(F_i) = 0$$

by Boole's inequality. (See Self-Test Exercise 14.)

Thus, with probability 1, the urn will be empty at 12 P.M. ■

7 Probability as a Measure of Belief

Thus far we have interpreted the probability of an event of a given experiment as being a measure of how frequently the event will occur when the experiment is continually repeated. However, there are also other uses of the term *probability*. For instance, we have all heard such statements as “It is 90 percent probable that Shakespeare actually wrote *Hamlet*” or “The probability that Oswald acted alone in assassinating Kennedy is .8.” How are we to interpret these statements?

The most simple and natural interpretation is that the probabilities referred to are measures of the individual's degree of belief in the statements that he or she is making. In other words, the individual making the foregoing statements is quite certain that Oswald acted alone and is even more certain that Shakespeare wrote *Hamlet*. This interpretation of probability as being a measure of the degree of one's belief is often referred to as the *personal* or *subjective* view of probability.

It seems logical to suppose that a “measure of the degree of one's belief” should satisfy all of the axioms of probability. For example, if we are 70 percent certain that Shakespeare wrote *Julius Caesar* and 10 percent certain that it was actually Marlowe, then it is logical to suppose that we are 80 percent certain that it was either

Shakespeare or Marlowe. Hence, whether we interpret probability as a measure of belief or as a long-run frequency of occurrence, its mathematical properties remain unchanged.

**Example
7a**

Suppose that in a 7-horse race, you believe that each of the first 2 horses has a 20 percent chance of winning, horses 3 and 4 each have a 15 percent chance, and the remaining 3 horses have a 10 percent chance each. Would it be better for you to wager at even money that the winner will be one of the first three horses or to wager, again at even money, that the winner will be one of the horses 1, 5, 6, and 7?

Solution On the basis of your personal probabilities concerning the outcome of the race, your probability of winning the first bet is $.2 + .2 + .15 = .55$, whereas it is $.2 + .1 + .1 + .1 = .5$ for the second bet. Hence, the first wager is more attractive. ■

Note that in supposing that a person's subjective probabilities are always consistent with the axioms of probability, we are dealing with an idealized rather than an actual person. For instance, if we were to ask someone what he thought the chances were of

- (a) rain today,
- (b) rain tomorrow,
- (c) rain both today and tomorrow,
- (d) rain either today or tomorrow,

it is quite possible that, after some deliberation, he might give 30 percent, 40 percent, 20 percent, and 60 percent as answers. Unfortunately, such answers (or such subjective probabilities) are not consistent with the axioms of probability. (Why not?) We would of course hope that after this was pointed out to the respondent, he would change his answers. (One possibility we could accept is 30 percent, 40 percent, 10 percent, and 60 percent.)

Summary

Let S denote the set of all possible outcomes of an experiment. S is called the *sample space* of the experiment. An *event* is a subset of S . If $A_i, i = 1, \dots, n$, are events, then $\bigcup_{i=1}^n A_i$, called the *union* of these events, consists of all outcomes that are in at least one of the events $A_i, i = 1, \dots, n$. Similarly, $\bigcap_{i=1}^n A_i$, sometimes written as $A_1 \cdots A_n$, is called the *intersection* of the events A_i and consists of all outcomes that are in all of the events $A_i, i = 1, \dots, n$.

For any event A , we define A^c to consist of all outcomes in the sample space that are not in A . We call A^c the *complement* of the event A . The event S^c , which is empty of outcomes, is designated by $\emptyset$ and is called the *null set*. If $AB = \emptyset$, then we say that A and B are *mutually exclusive*.

For each event A of the sample space S , we suppose that a number $P(A)$, called the probability of A , is defined and is such that

- (i) $0 \leq P(A) \leq 1$
- (ii) $P(S) = 1$
- (iii) For mutually exclusive events $A_i, i \geq 1$,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

$P(A)$ represents the *probability* that the outcome of the experiment is in A .

It can be shown that

$$P(A^c) = 1 - P(A)$$

A useful result is that

$$P(A \cup B) = P(A) + P(B) - P(AB)$$

which can be generalized to give

$$\begin{aligned} P\left(\bigcup_{i=1}^n A_i\right) &= \sum_{i=1}^n P(A_i) - \sum_{i < j} P(A_i A_j) \\ &\quad + \sum_{i < j < k} P(A_i A_j A_k) \\ &\quad + \cdots + (-1)^{n+1} P(A_1 \cdots A_n) \end{aligned}$$

This result is known as the *inclusion-exclusion identity*.

If S is finite and each one point set is assumed to have equal probability, then

$$P(A) = \frac{|A|}{|S|}$$

where $|E|$ denotes the number of outcomes in the event E . $P(A)$ can be interpreted either as a long-run relative frequency or as a measure of one's degree of belief.

Problems

1. A box contains 3 marbles: 1 red, 1 green, and 1 blue. Consider an experiment that consists of taking 1 marble from the box and then replacing it in the box and drawing a second marble from the box. Describe the sample space. Repeat when the second marble is drawn without replacing the first marble.

2. In an experiment, die is rolled continually until a 6 appears, at which point the experiment stops. What is the sample space of this experiment? Let E_n denote the event that n rolls are necessary to complete the experiment. What points of the sample space are contained in E_n ?

What is $\left(\bigcup_1^\infty E_n\right)^c$?

3. Two dice are thrown. Let E be the event that the sum of the dice is odd, let F be the event that at least one of the dice lands on 1, and let G be the event that the sum is 5. Describe the events EF , $E \cup F$, FG , EF^c , and EFG .

4. A , B , and C take turns flipping a coin. The first one to get a head wins. The sample space of this experiment can be defined by

$$S = \left\{ \begin{array}{l} 1, 01, 001, 0001, \dots, \\ 0000 \dots \end{array} \right.$$

(a) Interpret the sample space.

(b) Define the following events in terms of S :

(i) A wins = A .

(ii) B wins = B .

(iii) $(A \cup B)^c$.

Assume that A flips first, then B , then C , then A , and so on.

5. A system is composed of 5 components, each of which is either working or failed. Consider an experiment that consists of observing the status of each component, and let the outcome of the experiment be given by the vector $(x_1, x_2, x_3, x_4, x_5)$, where x_i is equal to 1 if component i is working and is equal to 0 if component i is failed.

(a) How many outcomes are in the sample space of this experiment?

(b) Suppose that the system will work if components 1 and 2 are both working, or if components 3 and 4 are both working, or if components 1, 3, and 5 are all working. Let W be the event that the system will work. Specify all the outcomes in W .

(c) Let A be the event that components 4 and 5 are both failed. How many outcomes are contained in the event A ?

(d) Write out all the outcomes in the event AW .

6. A hospital administrator codes incoming patients suffering gunshot wounds according to whether they have insurance (coding 1 if they do and 0 if they do not) and according to their condition, which is rated as good (g), fair (f), or serious (s). Consider an experiment that consists of the coding of such a patient.

(a) Give the sample space of this experiment.

(b) Let A be the event that the patient is in serious condition. Specify the outcomes in A .

(c) Let B be the event that the patient is uninsured. Specify the outcomes in B .

(d) Give all the outcomes in the event $B^c \cup A$.

7. Consider an experiment that consists of determining the type of job—either blue collar or white collar—and the political affiliation—Republican, Democratic, or Independent—of the 15 members of an adult soccer team. How many outcomes are

(a) in the sample space?

(b) in the event that at least one of the team members is a blue-collar worker?

(c) in the event that none of the team members considers himself or herself an Independent?

8. Suppose that A and B are mutually exclusive events for which $P(A) = .3$ and $P(B) = .5$. What is the probability that

(a) either A or B occurs?

(b) A occurs but B does not?

(c) both A and B occur?

9. A retail establishment accepts either the American Express or the VISA credit card. A total of 24 percent of its customers carry an American Express card, 61 percent carry a VISA card, and 11 percent carry both cards. What percentage of its customers carry a credit card that the establishment will accept?

10. Sixty percent of the students at a certain school wear neither a ring nor a necklace. Twenty percent wear a ring and 30 percent wear a necklace. If one of the students is

chosen randomly, what is the probability that this student is wearing

- (a) a ring or a necklace?
- (b) a ring and a necklace?

11. A total of 28 percent of American males smoke cigarettes, 7 percent smoke cigars, and 5 percent smoke both cigars and cigarettes.

- (a) What percentage of males smokes neither cigars nor cigarettes?
- (b) What percentage smokes cigars but not cigarettes?

12. An elementary school is offering 3 language classes: one in Spanish, one in French, and one in German. The classes are open to any of the 100 students in the school. There are 28 students in the Spanish class, 26 in the French class, and 16 in the German class. There are 12 students who are in both Spanish and French, 4 who are in both Spanish and German, and 6 who are in both French and German. In addition, there are 2 students taking all 3 classes.

- (a) If a student is chosen randomly, what is the probability that he or she is not in any of the language classes?
- (b) If a student is chosen randomly, what is the probability that he or she is taking exactly one language class?
- (c) If 2 students are chosen randomly, what is the probability that at least 1 is taking a language class?

13. A certain town with a population of 100,000 has 3 newspapers: I, II, and III. The proportions of townspeople who read these papers are as follows:

I: 10 percent I and II: 8 percent I and II and III: 1 percent
 II: 30 percent I and III: 2 percent
 III: 5 percent II and III: 4 percent

(The list tells us, for instance, that 8000 people read newspapers I and II.)

- (a) Find the number of people who read only one newspaper.
- (b) How many people read at least two newspapers?
- (c) If I and III are morning papers and II is an evening paper, how many people read at least one morning paper plus an evening paper?
- (d) How many people do not read any newspapers?
- (e) How many people read only one morning paper and one evening paper?

14. The following data were given in a study of a group of 1000 subscribers to a certain magazine: In reference to job, marital status, and education, there were 312 professionals, 470 married persons, 525 college graduates, 42 professional college graduates, 147 married college graduates, 86 married professionals, and 25 married professional college graduates. Show that the numbers reported in the study must be incorrect.

Hint: Let M , W , and G denote, respectively, the set of professionals, married persons, and college graduates. Assume that one of the 1000 persons is chosen at random, and use Proposition 4.4 to show that if the given numbers are correct, then $P(M \cup W \cup G) > 1$.

15. If it is assumed that all $\binom{52}{5}$ poker hands are equally likely, what is the probability of being dealt

- (a) a flush? (A hand is said to be a flush if all 5 cards are of the same suit.)
- (b) one pair? (This occurs when the cards have denominations a, a, b, c, d , where a, b, c , and d are all distinct.)
- (c) two pairs? (This occurs when the cards have denominations a, a, b, b, c , where a, b , and c are all distinct.)
- (d) three of a kind? (This occurs when the cards have denominations a, a, a, b, c , where a, b , and c are all distinct.)
- (e) four of a kind? (This occurs when the cards have denominations a, a, a, a, b .)

16. Poker dice is played by simultaneously rolling 5 dice. Show that

- (a) $P\{\text{no two alike}\} = .0926$;
- (b) $P\{\text{one pair}\} = .4630$;
- (c) $P\{\text{two pair}\} = .2315$;
- (d) $P\{\text{three alike}\} = .1543$;
- (e) $P\{\text{full house}\} = .0386$;
- (f) $P\{\text{four alike}\} = .0193$;
- (g) $P\{\text{five alike}\} = .0008$.

17. If 8 rooks (castles) are randomly placed on a chessboard, compute the probability that none of the rooks can capture any of the others. That is, compute the probability that no row or file contains more than one rook.

18. Two cards are randomly selected from an ordinary playing deck. What is the probability that they form a blackjack? That is, what is the probability that one of the cards is an ace and the other one is either a ten, a jack, a queen, or a king?

19. Two symmetric dice have had two of their sides painted red, two painted black, one painted yellow, and the other painted white. When this pair of dice is rolled, what is the probability that both dice land with the same color face up?

20. Suppose that you are playing blackjack against a dealer. In a freshly shuffled deck, what is the probability that neither you nor the dealer is dealt a blackjack?

21. A small community organization consists of 20 families, of which 4 have one child, 8 have two children, 5

have three children, 2 have four children, and 1 has five children.

(a) If one of these families is chosen at random, what is the probability it has i children, $i = 1, 2, 3, 4, 5$?

(b) If one of the children is randomly chosen, what is the probability that child comes from a family having i children, $i = 1, 2, 3, 4, 5$?

22. Consider the following technique for shuffling a deck of n cards: For any initial ordering of the cards, go through the deck one card at a time and at each card, flip a fair coin. If the coin comes up heads, then leave the card where it is; if the coin comes up tails, then move that card to the end of the deck. After the coin has been flipped n times, say that one round has been completed. For instance, if $n = 4$ and the initial ordering is 1, 2, 3, 4, then if the successive flips result in the outcome h, t, t, h , then the ordering at the end of the round is 1, 4, 2, 3. Assuming that all possible outcomes of the sequence of n coin flips are equally likely, what is the probability that the ordering after one round is the same as the initial ordering?

23. A pair of fair dice is rolled. What is the probability that the second die lands on a higher value than does the first?

24. If two dice are rolled, what is the probability that the sum of the upturned faces equals i ? Find it for $i = 2, 3, \dots, 11, 12$.

25. A pair of dice is rolled until a sum of either 5 or 7 appears. Find the probability that a 5 occurs first.

Hint: Let E_n denote the event that a 5 occurs on the n th roll and no 5 or 7 occurs on the first $n - 1$ rolls. Compute $P(E_n)$ and argue that $\sum_{n=1}^{\infty} P(E_n)$ is the desired probability.

26. The game of craps is played as follows: A player rolls two dice. If the sum of the dice is either a 2, 3, or 12, the player loses; if the sum is either a 7 or an 11, the player wins. If the outcome is anything else, the player continues to roll the dice until she rolls either the initial outcome or a 7. If the 7 comes first, the player loses, whereas if the initial outcome reoccurs before the 7 appears, the player wins. Compute the probability of a player winning at craps.

Hint: Let E_i denote the event that the initial outcome is i and the player wins. The desired probability is $\sum_{i=2}^{12} P(E_i)$.

To compute $P(E_i)$, define the events $E_{i,n}$ to be the event that the initial sum is i and the player wins on the n th roll.

Argue that $P(E_i) = \sum_{n=1}^{\infty} P(E_{i,n})$.

27. An urn contains 3 red and 7 black balls. Players A and B withdraw balls from the urn consecutively until a red ball is selected. Find the probability that A selects the red ball. (A draws the first ball, then B , and so on. There is no replacement of the balls drawn.)

28. An urn contains 5 red, 6 blue, and 8 green balls. If a set of 3 balls is randomly selected, what is the probability that each of the balls will be (a) of the same color? (b) of different colors? Repeat under the assumption that whenever a ball is selected, its color is noted and it is then replaced in the urn before the next selection. This is known as *sampling with replacement*.

29. An urn contains n white and m black balls, where n and m are positive numbers.

(a) If two balls are randomly withdrawn, what is the probability that they are the same color?

(b) If a ball is randomly withdrawn and then replaced before the second one is drawn, what is the probability that the withdrawn balls are the same color?

(c) Show that the probability in part (b) is always larger than the one in part (a).

30. The chess clubs of two schools consist of, respectively, 8 and 9 players. Four members from each club are randomly chosen to participate in a contest between the two schools. The chosen players from one team are then randomly paired with those from the other team, and each pairing plays a game of chess. Suppose that Rebecca and her sister Elise are on the chess clubs at different schools. What is the probability that

(a) Rebecca and Elise will be paired?

(b) Rebecca and Elise will be chosen to represent their schools but will not play each other?

(c) either Rebecca or Elise will be chosen to represent her school?

31. A 3-person basketball team consists of a guard, a forward, and a center.

(a) If a person is chosen at random from each of three different such teams, what is the probability of selecting a complete team?

(b) What is the probability that all 3 players selected play the same position?

32. A group of individuals containing b boys and g girls is lined up in random order; that is, each of the $(b + g)!$ permutations is assumed to be equally likely. What is the probability that the person in the i th position, $1 \leq i \leq b + g$, is a girl?

33. A forest contains 20 elk, of which 5 are captured, tagged, and then released. A certain time later, 4 of the 20 elk are captured. What is the probability that 2 of these 4 have been tagged? What assumptions are you making?

34. The second Earl of Yarborough is reported to have bet at odds of 1000 to 1 that a bridge hand of 13 cards would contain at least one card that is ten or higher. (By *ten or higher* we mean that a card is either a ten, a jack, a queen, a king, or an ace.) Nowadays, we call a hand that has no cards higher than 9 a *Yarborough*. What is the probability that a randomly selected bridge hand is a Yarborough?

35. Seven balls are randomly withdrawn from an urn that contains 12 red, 16 blue, and 18 green balls. Find the probability that

- (a) 3 red, 2 blue, and 2 green balls are withdrawn;
- (b) at least 2 red balls are withdrawn;
- (c) all withdrawn balls are the same color;
- (d) either exactly 3 red balls or exactly 3 blue balls are withdrawn.

36. Two cards are chosen at random from a deck of 52 playing cards. What is the probability that they

- (a) are both aces?
- (b) have the same value?

37. An instructor gives her class a set of 10 problems with the information that the final exam will consist of a random selection of 5 of them. If a student has figured out how to do 7 of the problems, what is the probability that he or she will answer correctly

- (a) all 5 problems?
- (b) at least 4 of the problems?

38. There are n socks, 3 of which are red, in a drawer. What is the value of n if, when 2 of the socks are chosen randomly, the probability that they are both red is $\frac{1}{2}$?

39. There are 5 hotels in a certain town. If 3 people check into hotels in a day, what is the probability that they each check into a different hotel? What assumptions are you making?

40. A town contains 4 people who repair televisions. If 4 sets break down, what is the probability that exactly i of the repairers are called? Solve the problem for $i = 1, 2, 3, 4$. What assumptions are you making?

41. If a die is rolled 4 times, what is the probability that 6 comes up at least once?

42. Two dice are thrown n times in succession. Compute the probability that double 6 appears at least once. How large need n be to make this probability at least $\frac{1}{2}$?

43. (a) If N people, including A and B , are randomly arranged in a line, what is the probability that A and B are next to each other?

(b) What would the probability be if the people were randomly arranged in a circle?

44. Five people, designated as A, B, C, D, E , are arranged in linear order. Assuming that each possible order is equally likely, what is the probability that

- (a) there is exactly one person between A and B ?
- (b) there are exactly two people between A and B ?
- (c) there are three people between A and B ?

45. A woman has n keys, of which one will open her door.

(a) If she tries the keys at random, discarding those that do not work, what is the probability that she will open the door on her k th try?

(b) What if she does not discard previously tried keys?

46. How many people have to be in a room in order that the probability that at least two of them celebrate their birthday in the same month is at least $\frac{1}{2}$? Assume that all possible monthly outcomes are equally likely.

47. If there are 12 strangers in a room, what is the probability that no two of them celebrate their birthday in the same month?

48. Given 20 people, what is the probability that among the 12 months in the year, there are 4 months containing exactly 2 birthdays and 4 containing exactly 3 birthdays?

49. A group of 6 men and 6 women is randomly divided into 2 groups of size 6 each. What is the probability that both groups will have the same number of men?

50. In a hand of bridge, find the probability that you have 5 spades and your partner has the remaining 8.

51. Suppose that n balls are randomly distributed into N compartments. Find the probability that m balls will fall into the first compartment. Assume that all N^n arrangements are equally likely.

52. A closet contains 10 pairs of shoes. If 8 shoes are randomly selected, what is the probability that there will be

- (a) no complete pair?
- (b) exactly 1 complete pair?

53. If 4 married couples are arranged in a row, find the probability that no husband sits next to his wife.

54. Compute the probability that a bridge hand is void in at least one suit. Note that the answer is not

$$\frac{\binom{4}{1} \binom{39}{13}}{\binom{52}{13}}$$

(Why not?)

Hint: Use Proposition 4.4.

55. Compute the probability that a hand of 13 cards contains

- (a) the ace and king of at least one suit;
- (b) all 4 of at least 1 of the 13 denominations.

56. Two players play the following game: Player A chooses one of the three spinners pictured in Figure 6, and then player B chooses one of the remaining two spinners. Both players then spin their spinner, and the one that lands on the higher number is declared the winner. Assuming that each spinner is equally likely to land in any of its 3 regions, would you rather be player A or player B ? Explain your answer!

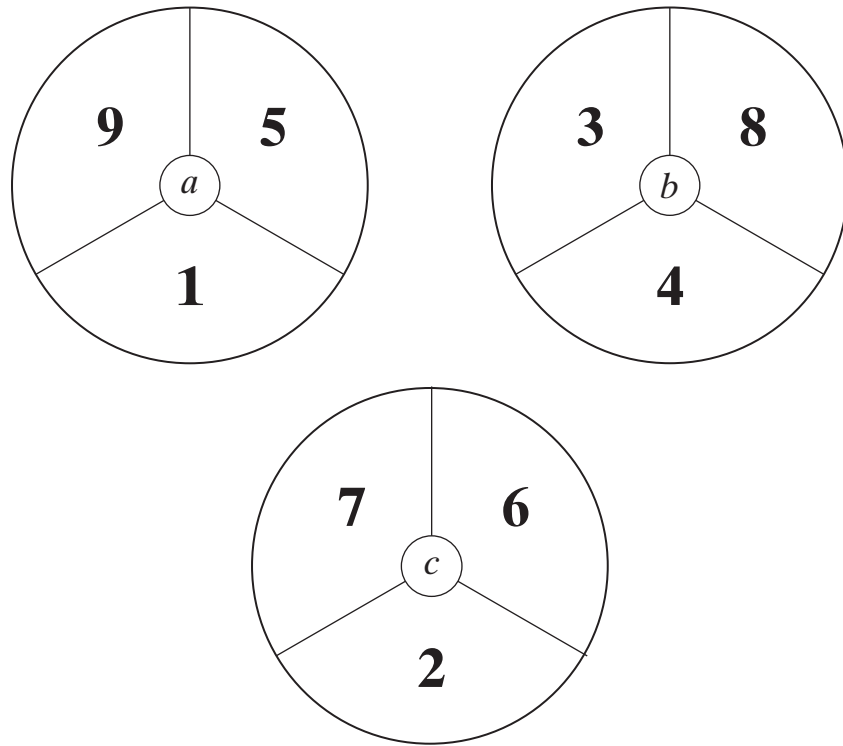


Figure 6 Spinners.

Theoretical Exercises

Prove the following relations:

1. $EF \subset E \subset E \cup F$.

2. If $E \subset F$, then $F^c \subset E^c$.

3. $F = FE \cup FE^c$ and $E \cup F = E \cup E^c F$.

4. $\left(\bigcup_{i=1}^{\infty} E_i \right) F = \bigcup_{i=1}^{\infty} E_i F$ and

$\left(\bigcap_{i=1}^{\infty} E_i \right) \cup F = \bigcap_{i=1}^{\infty} (E_i \cup F)$.

5. For any sequence of events $E_1, E_2, \dots$, define a new sequence $F_1, F_2, \dots$ of disjoint events (that is, events such that $F_i F_j = \emptyset$ whenever $i \neq j$) such that for all $n \geq 1$,

$$\bigcup_{i=1}^n F_i = \bigcup_{i=1}^n E_i$$

6. Let E, F , and G be three events. Find expressions for the events so that, of E, F , and G ,

- (a) only E occurs;
- (b) both E and G , but not F , occur;
- (c) at least one of the events occurs;
- (d) at least two of the events occur;
- (e) all three events occur;
- (f) none of the events occurs;
- (g) at most one of the events occurs;
- (h) at most two of the events occur;
- (i) exactly two of the events occur;
- (j) at most three of the events occur.

7. Use Venn diagrams

- (a) to simplify the expressions $(E \cup F)(E \cup F^c)$;
- (b) to prove DeMorgan's laws for events E and F . [That is, prove $(E \cup F)^c = E^c F^c$, and $(EF)^c = E^c \cup F^c$.]

8. Let S be a given set. If, for some $k > 0$, $S_1, S_2, \dots, S_k$ are mutually exclusive nonempty subsets of S such that

$\bigcup_{i=1}^k S_i = S$, then we call the set $\{S_1, S_2, \dots, S_k\}$ a *partition* of S . Let T_n denote the number of different partitions of $\{1, 2, \dots, n\}$. Thus, $T_1 = 1$ (the only partition being $S_1 = \{1\}$) and $T_2 = 2$ (the two partitions being $\{\{1, 2\}\}, \{\{1\}, \{2\}\}$).

- (a) Show, by computing all partitions, that $T_3 = 5, T_4 = 15$.
 (b) Show that

$$T_{n+1} = 1 + \sum_{k=1}^n \binom{n}{k} T_k$$

and use this equation to compute T_{10} .

Hint: One way of choosing a partition of $n + 1$ items is to call one of the items *special*. Then we obtain different partitions by first choosing $k, k = 0, 1, \dots, n$, then a subset of size $n - k$ of the nonspecial items, and then any of the T_k partitions of the remaining k nonspecial items. By adding the special item to the subset of size $n - k$, we obtain a partition of all $n + 1$ items.

9. Suppose that an experiment is performed n times. For any event E of the sample space, let $n(E)$ denote the number of times that event E occurs and define $f(E) = n(E)/n$. Show that $f(\cdot)$ satisfies Axioms 1, 2, and 3.

10. Prove that $P(E \cup F \cup G) = P(E) + P(F) + P(G) - P(E^c F G) - P(E F^c G) - P(E F G^c) - 2P(E F G)$.

11. If $P(E) = .9$ and $P(F) = .8$, show that $P(EF) \geq .7$. In general, prove *Bonferroni's inequality*, namely,

$$P(EF) \geq P(E) + P(F) - 1$$

12. Show that the probability that exactly one of the events E or F occurs equals $P(E) + P(F) - 2P(EF)$.

13. Prove that $P(EF^c) = P(E) - P(EF)$.

14. Prove Proposition 4.4 by mathematical induction.

15. An urn contains M white and N black balls. If a random sample of size r is chosen, what is the probability that it contains exactly k white balls?

16. Use induction to generalize *Bonferroni's inequality* to n events. That is, show that

$$P(E_1 E_2 \cdots E_n) \geq P(E_1) + \cdots + P(E_n) - (n - 1)$$

17. Consider the matching problem, Example 5m, and define A_N to be the number of ways in which the N men can select their hats so that no man selects his own. Argue that

$$A_N = (N - 1)(A_{N-1} + A_{N-2})$$

This formula, along with the boundary conditions $A_1 = 0, A_2 = 1$, can then be solved for A_N , and the desired probability of no matches would be $A_N/N!$.

Hint: After the first man selects a hat that is not his own, there remain $N - 1$ men to select among a set of $N - 1$ hats that does not contain the hat of one of these men. Thus, there is one extra man and one extra hat. Argue that we can get no matches either with the extra man selecting the extra hat or with the extra man not selecting the extra hat.

18. Let f_n denote the number of ways of tossing a coin n times such that successive heads never appear. Argue that

$$f_n = f_{n-1} + f_{n-2} \quad n \geq 2, \text{ where } f_0 = 1, f_1 = 2$$

Hint: How many outcomes are there that start with a head, and how many start with a tail? If P_n denotes the probability that successive heads never appear when a coin is tossed n times, find P_n (in terms of f_n) when all possible outcomes of the n tosses are assumed equally likely. Compute P_{10} .

19. An urn contains n red and m blue balls. They are withdrawn one at a time until a total of $r, r \leq n$, red balls have been withdrawn. Find the probability that a total of k balls are withdrawn.

Hint: A total of k balls will be withdrawn if there are $r - 1$ red balls in the first $k - 1$ withdrawals and the k th withdrawal is a red ball.

20. Consider an experiment whose sample space consists of a countably infinite number of points. Show that not all points can be equally likely. Can all points have a positive probability of occurring?

- *21. Consider Example 5o, which is concerned with the number of runs of wins obtained when n wins and m losses are randomly permuted. Now consider the total number of runs—that is, win runs plus loss runs—and show that

$$\begin{aligned} P\{2k \text{ runs}\} &= 2 \frac{\binom{m-1}{k-1} \binom{n-1}{k-1}}{\binom{m+n}{n}} \\ P\{2k+1 \text{ runs}\} &= \frac{\binom{m-1}{k-1} \binom{n-1}{k} + \binom{m-1}{k} \binom{n-1}{k-1}}{\binom{m+n}{n}} \end{aligned}$$

Self-Test Problems and Exercises

1. A cafeteria offers a three-course meal consisting of an entree, a starch, and a dessert. The possible choices are given in the following table:

Course	Choices
Entree	Chicken or roast beef
Starch	Pasta or rice or potatoes
Dessert	Ice cream or Jello or apple pie or a peach

A person is to choose one course from each category.

- (a) How many outcomes are in the sample space?
- (b) Let A be the event that ice cream is chosen. How many outcomes are in A ?
- (c) Let B be the event that chicken is chosen. How many outcomes are in B ?
- (d) List all the outcomes in the event AB .
- (e) Let C be the event that rice is chosen. How many outcomes are in C ?
- (f) List all the outcomes in the event ABC .

2. A customer visiting the suit department of a certain store will purchase a suit with probability .22, a shirt with probability .30, and a tie with probability .28. The customer will purchase both a suit and a shirt with probability .11, both a suit and a tie with probability .14, and both a shirt and a tie with probability .10. A customer will purchase all 3 items with probability .06. What is the probability that a customer purchases

- (a) none of these items?
- (b) exactly 1 of these items?

3. A deck of cards is dealt out. What is the probability that the 14th card dealt is an ace? What is the probability that the first ace occurs on the 14th card?

4. Let A denote the event that the midtown temperature in Los Angeles is 70°F , and let B denote the event that the midtown temperature in New York is 70°F . Also, let C denote the event that the maximum of the midtown temperatures in New York and in Los Angeles is 70°F . If $P(A) = .3$, $P(B) = .4$, and $P(C) = .2$, find the probability that the minimum of the two midtown temperatures is 70°F .

5. An ordinary deck of 52 cards is shuffled. What is the probability that the top four cards have

- (a) different denominations?
- (b) different suits?

6. Urn A contains 3 red and 3 black balls, whereas urn B contains 4 red and 6 black balls. If a ball is randomly

selected from each urn, what is the probability that the balls will be the same color?

7. In a state lottery, a player must choose 8 of the numbers from 1 to 40. The lottery commission then performs an experiment that selects 8 of these 40 numbers. Assuming that the choice of the lottery commission is equally likely to be any of the $\binom{40}{8}$ combinations, what is the probability that a player has

- (a) all 8 of the numbers selected by the lottery commission?
- (b) 7 of the numbers selected by the lottery commission?
- (c) at least 6 of the numbers selected by the lottery commission?

8. From a group of 3 first-year students, 4 sophomores, 4 juniors, and 3 seniors, a committee of size 4 is randomly selected. Find the probability that the committee will consist of

- (a) 1 from each class;
- (b) 2 sophomores and 2 juniors;
- (c) only sophomores or juniors.

9. For a finite set A , let $N(A)$ denote the number of elements in A .

- (a) Show that

$$N(A \cup B) = N(A) + N(B) - N(AB)$$

- (b) More generally, show that

$$N\left(\bigcup_{i=1}^n A_i\right) = \sum_i N(A_i) - \sum_{i < j} N(A_i A_j) + \cdots + (-1)^{n+1} N(A_1 \cdots A_n)$$

10. Consider an experiment that consists of 6 horses, numbered 1 through 6, running a race, and suppose that the sample space consists of the $6!$ possible orders in which the horses finish. Let A be the event that the number-1 horse is among the top three finishers, and let B be the event that the number-2 horse comes in second. How many outcomes are in the event $A \cup B$?

11. A 5-card hand is dealt from a well-shuffled deck of 52 playing cards. What is the probability that the hand contains at least one card from each of the four suits?

12. A basketball team consists of 6 frontcourt and 4 backcourt players. If players are divided into roommates at random, what is the probability that there will be exactly two roommate pairs made up of a backcourt and a frontcourt player?

13. Suppose that a person chooses a letter at random from R E S E R V E and then chooses one at random from V E R T I C A L. What is the probability that the same letter is chosen?

14. Prove *Boole's inequality*:

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} P(A_i)$$

15. Show that if $P(A_i) = 1$ for all $i \geq 1$, then $P\left(\bigcap_{i=1}^{\infty} A_i\right) = 1$.

16. Let $T_k(n)$ denote the number of partitions of the set $\{1, \dots, n\}$ into k nonempty subsets, where $1 \leq k \leq n$. (See Theoretical Exercise 8 for the definition of a partition.) Argue that

$$T_k(n) = kT_k(n-1) + T_{k-1}(n-1)$$

Hint: In how many partitions is $\{1\}$ a subset, and in how many is 1 an element of a subset that contains other elements?

17. Five balls are randomly chosen, without replacement, from an urn that contains 5 red, 6 white, and 7 blue balls.

Find the probability that at least one ball of each color is chosen.

18. Four red, 8 blue, and 5 green balls are randomly arranged in a line.

(a) What is the probability that the first 5 balls are blue?

(b) What is the probability that none of the first 5 balls is blue?

(c) What is the probability that the final 3 balls are of different colors?

(d) What is the probability that all the red balls are together?

19. Ten cards are randomly chosen from a deck of 52 cards that consists of 13 cards of each of 4 different suits. Each of the selected cards is put in one of 4 piles, depending on the suit of the card.

(a) What is the probability that the largest pile has 4 cards, the next largest has 3, the next largest has 2, and the smallest has 1 card?

(b) What is the probability that two of the piles have 3 cards, one has 4 cards, and one has no cards?

20. Balls are randomly removed from an urn initially containing 20 red and 10 blue balls. What is the probability that all of the red balls are removed before all of the blue ones have been removed?

Answers to Selected Problems

9. 74 **10.** .4; .1 **11.** 70; 2 **12.** .5; .32; 149/198
13. 20,000; 12,000; 11,000; 68,000; 10,000 **14.** 1.057
15. .0020; .4226; .0475; .0211; .00024 **17.** 9.10947×10^{-6}
18. .048 **19.** 5/18 **20.** .9052 **22.** $(n+1)/2^n$ **23.** 5/12
25. .4 **26.** .492929 **28.** .0888; .2477; .1243; .2099
30. 1/18; 1/6; 1/2 **31.** 2/9; 1/9 **33.** 70/323 **34.** 1001;
120; 495 **36.** .0045; .0588 **37.** .0833; .5 **38.** 4 **39.** .48
40. 1/64; 21/64; 36/64; 6/64 **41.** .5177 **44.** .3; .2;
.1 **46.** 5 **48.** .01697 **49.** .4329 **50.** 2.6084×10^{-6}
52. .09145; .4268 **53.** 12/35 **54.** .0511 **55.** .2198;
.0342

Solutions to Self-Test Problems and Exercises

1. (a) $2 \cdot 3 \cdot 4 = 24$

(b) $2 \cdot 3 = 6$

(c) $3 \cdot 4 = 12$

(d) $AB = \{(c, \text{pasta}, i), (c, \text{rice}, i), (c, \text{potatoes}, i)\}$

(e) 8

(f) $ABC = \{(c, \text{rice}, i)\}$

2. Let A be the event that a suit is purchased, B be the event that a shirt is purchased, and C be the event that a tie is purchased. Then

$$P(A \cup B \cup C) = .22 + .30 + .28 - .11 - .14 - .10 + .06 = .51$$

(a) $1 - .51 = .49$

(b) The probability that two or more items are purchased is

$$P(AB \cup AC \cup BC) = .11 + .14 + .10 - .06 - .06 - .06 + .06 = .23$$

Hence, the probability that exactly 1 item is purchased is $.51 - .23 = .28$.

3. By symmetry, the 14th card is equally likely to be any of the 52 cards; thus, the probability is $4/52$. A more formal argument is to count the number of the $52!$ outcomes for which the 14th card is an ace. This yields

$$p = \frac{4 \cdot 51 \cdot 50 \cdots 2 \cdot 1}{(52)!} = \frac{4}{52}$$

Letting A be the event that the first ace occurs on the 14th card, we have

$$P(A) = \frac{48 \cdot 47 \cdots 36 \cdot 4}{52 \cdot 51 \cdots 40 \cdot 39} = .0312$$

4. Let D denote the event that the minimum temperature is 70 degrees. Then

$$P(A \cup B) = P(A) + P(B) - P(AB) = .7 - P(AB)$$

$$P(C \cup D) = P(C) + P(D) - P(CD) = .2 + P(D) - P(DC)$$

Since $A \cup B = C \cup D$ and $AB = CD$, subtracting one of the preceding equations from the other yields

$$0 = .5 - P(D)$$

or $P(D) = .5$.

$$5. (a) \frac{52 \cdot 48 \cdot 44 \cdot 40}{52 \cdot 51 \cdot 50 \cdot 49} = .6761$$

$$(b) \frac{52 \cdot 39 \cdot 26 \cdot 13}{52 \cdot 51 \cdot 50 \cdot 49} = .1055$$

6. Let R be the event that both balls are red, and let B be the event that both are black. Then

$$P(R \cup B) = P(R) + P(B) = \frac{3 \cdot 4}{6 \cdot 10} + \frac{3 \cdot 6}{6 \cdot 10} = 1/2$$

$$7. (a) \frac{1}{\binom{40}{8}} = 1.3 \times 10^{-8}$$

$$(b) \frac{\binom{8}{7} \binom{32}{1}}{\binom{40}{8}} = 3.3 \times 10^{-6}$$

$$(c) \frac{\binom{8}{6} \binom{32}{2}}{\binom{40}{8}} + 1.3 \times 10^{-8} + 3.3 \times 10^{-6} = 1.8 \times 10^{-4}$$

$$8. (a) \frac{3 \cdot 4 \cdot 4 \cdot 3}{\binom{14}{4}} = .1439$$

$$(b) \frac{\binom{4}{2} \binom{4}{2}}{\binom{14}{4}} = .0360$$

$$(c) \frac{\binom{8}{4}}{\binom{14}{4}} = .0699$$

9. Let $S = \bigcup_{i=1}^n A_i$, and consider the experiment of randomly choosing an element of S . Then $P(A) = N(A)/N(S)$.

10. Since there are $5! = 120$ outcomes in which the position of horse number 1 is specified, it follows that $N(A) = 360$. Similarly, $N(B) = 120$, and $N(AB) = 2 \cdot 4! = 48$. Hence, from Self-Test Problem 9, we obtain $N(A \cup B) = 432$.

11. One way to solve this problem is to start with the complementary probability that at least one suit does not appear. Let A_i , $i = 1, 2, 3, 4$, be the event that no cards from suit i appear. Then

$$\begin{aligned} P\left(\bigcup_{i=1}^4 A_i\right) &= \sum_i P(A_i) - \sum_j \sum_{i:i < j} P(A_i A_j) \\ &\quad + \cdots - P(A_1 A_2 A_3 A_4) \\ &= 4 \frac{\binom{39}{5}}{\binom{52}{5}} - \binom{4}{2} \frac{\binom{26}{5}}{\binom{52}{5}} + \binom{4}{3} \frac{\binom{13}{5}}{\binom{52}{5}} \\ &= 4 \frac{\binom{39}{5}}{\binom{52}{5}} - 6 \frac{\binom{26}{5}}{\binom{52}{5}} + 4 \frac{\binom{13}{5}}{\binom{52}{5}} \end{aligned}$$

The desired probability is then 1 minus the preceding. Another way to solve is to let A be the event that all 4 suits are represented, and then use

$$P(A) = P(n, n, n, n, o) + P(n, n, n, o, n) + P(n, n, o, n, n) + P(n, o, n, n, n)$$

where $P(n, n, n, o, n)$, for instance, is the probability that the first card is from a new suit, the second is from a new suit, the third is from a new suit, the fourth is from an old suit (that is, one which has already appeared) and the fifth is from a new suit. This gives

$$\begin{aligned} P(A) &= \frac{52 \cdot 39 \cdot 26 \cdot 13 \cdot 48 + 52 \cdot 39 \cdot 26 \cdot 36 \cdot 13}{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48} \\ &\quad + \frac{52 \cdot 39 \cdot 24 \cdot 26 \cdot 13 + 52 \cdot 12 \cdot 39 \cdot 26 \cdot 13}{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48} \\ &= \frac{52 \cdot 39 \cdot 26 \cdot 13(48 + 36 + 24 + 12)}{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48} \\ &= .2637 \end{aligned}$$

12. There are $(10)!/2^5$ different divisions of the 10 players into a first roommate pair, a second roommate pair, and so on. Hence, there are $(10)!/(5!2^5)$ divisions into 5 roommate pairs. There are $\binom{6}{2} \binom{4}{2}$ ways of choosing the frontcourt and backcourt players to be in the mixed roommate pairs and then 2 ways of pairing them up. As there is

then 1 way to pair up the remaining two backcourt players and $4!/(2!2!) = 3$ ways of making two roommate pairs from the remaining four frontcourt players, the desired probability is

$$P\{2 \text{ mixed pairs}\} = \frac{\binom{6}{2} \binom{4}{2} (2)(3)}{(10)!/(5!2^5)} = .5714$$

13. Let R denote the event that letter R is repeated; similarly, define the events E and V . Then

$$\begin{aligned} P\{\text{same letter}\} &= P(R) + P(E) + P(V) \\ &= \frac{2}{7} \frac{1}{8} + \frac{3}{7} \frac{1}{8} + \frac{1}{7} \frac{1}{8} = \frac{3}{28} \end{aligned}$$

14. Let $B_1 = A_1, B_i = A_i \left(\bigcup_{j=1}^{i-1} A_j \right)^c, i > 1$. Then

$$\begin{aligned} P\left(\bigcup_{i=1}^{\infty} A_i\right) &= P\left(\bigcup_{i=1}^{\infty} B_i\right) \\ &= \sum_{i=1}^{\infty} P(B_i) \\ &\leq \sum_{i=1}^{\infty} P(A_i) \end{aligned}$$

where the final equality uses the fact that the B_i are mutually exclusive. The inequality then follows, since $B_i \subset A_i$.

15.

$$\begin{aligned} P\left(\bigcap_{i=1}^{\infty} A_i\right) &= 1 - P\left(\left(\bigcap_{i=1}^{\infty} A_i\right)^c\right) \\ &= 1 - P\left(\bigcup_{i=1}^{\infty} A_i^c\right) \\ &\geq 1 - \sum_{i=1}^{\infty} P(A_i^c) \\ &= 1 \end{aligned}$$

16. The number of partitions for which $\{1\}$ is a subset is equal to the number of partitions of the remaining $n - 1$ elements into $k - 1$ nonempty subsets, namely, $T_{k-1}(n - 1)$. Because there are $T_k(n - 1)$ partitions of $\{2, \dots, n - 1\}$ into k nonempty subsets and then a choice of k of them in which to place element 1, it follows that there are $kT_k(n - 1)$ partitions for which $\{1\}$ is not a subset. Hence, the result follows.

17. Let R, W, B denote, respectively, the events that there are no red, no white, and no blue balls chosen. Then

$$\begin{aligned} P(R \cup W \cup B) &= P(R) + P(W) + P(B) - P(RW) \\ &\quad - P(RB) - P(WB) + P(RWB) \\ &= \frac{\binom{13}{5}}{\binom{18}{5}} + \frac{\binom{12}{5}}{\binom{18}{5}} + \frac{\binom{11}{5}}{\binom{18}{5}} - \frac{\binom{7}{5}}{\binom{18}{5}} \\ &\quad - \frac{\binom{6}{5}}{\binom{18}{5}} - \frac{\binom{5}{5}}{\binom{18}{5}} \\ &\approx 0.2933 \end{aligned}$$

Thus, the probability that all colors appear in the chosen subset is approximately $1 - 0.2933 = 0.7067$.

18. (a) $\frac{8 \cdot 7 \cdot 6 \cdot 5 \cdot 4}{17 \cdot 16 \cdot 15 \cdot 14 \cdot 13} = \frac{2}{221}$

(b) Because there are 9 nonblue balls, the probability is $\frac{9 \cdot 8 \cdot 7 \cdot 6 \cdot 5}{17 \cdot 16 \cdot 15 \cdot 14 \cdot 13} = \frac{9}{442}$.

(c) Because there are $3!$ possible orderings of the different colors and all possibilities for the final 3 balls are equally likely, the probability is $\frac{3! \cdot 4 \cdot 8 \cdot 5}{17 \cdot 16 \cdot 15} = \frac{4}{17}$.

(d) The probability that the red balls are in a specified 4 spots is $\frac{4 \cdot 3 \cdot 2 \cdot 1}{17 \cdot 16 \cdot 15 \cdot 14}$. Because there are 14 possible locations of the red balls where they are all together, the probability is $\frac{14 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{17 \cdot 16 \cdot 15 \cdot 14} = \frac{1}{170}$.

19. (a) The probability that the 10 cards consist of 4 spades, 3 hearts, 2 diamonds, and 1 club is $\frac{\binom{13}{4} \binom{13}{3} \binom{13}{2} \binom{13}{1}}{\binom{52}{10}}$. Because there are $4!$ possible choices of the suits to have 4, 3, 2, and 1 cards, respectively, it follows that the probability is $\frac{24 \binom{13}{4} \binom{13}{3} \binom{13}{2} \binom{13}{1}}{\binom{52}{10}}$.

(b) Because there are $\binom{4}{2} = 6$ choices of the two suits that are to have 3 cards and then 2 choices for the suit to have 4 cards, the probability is $\frac{12 \binom{13}{3} \binom{13}{3} \binom{13}{4}}{\binom{52}{10}}$.

20. All the red balls are removed before all the blue ones if and only if the very last ball removed is blue. Because all 30 balls are equally likely to be the last ball removed, the probability is $10/30$.